



CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

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I. Record of Changes

Date	A* M, D	In charge	Change Description
12/06/2024	A	Nguyen Duc Anh	Create report item(1,2,3)

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Overall Requirements

1.1 Context Diagram

The Medpal System is a website that replaces the current manual and telephone processes for booking appointments and accessing patient information at **68A polyclinic**. The context diagram below represents the system's components, subsequent connections and communications, workflows and processes within the system. This diagram provides important information about how software can improve the connection between the polyclinic and patients.

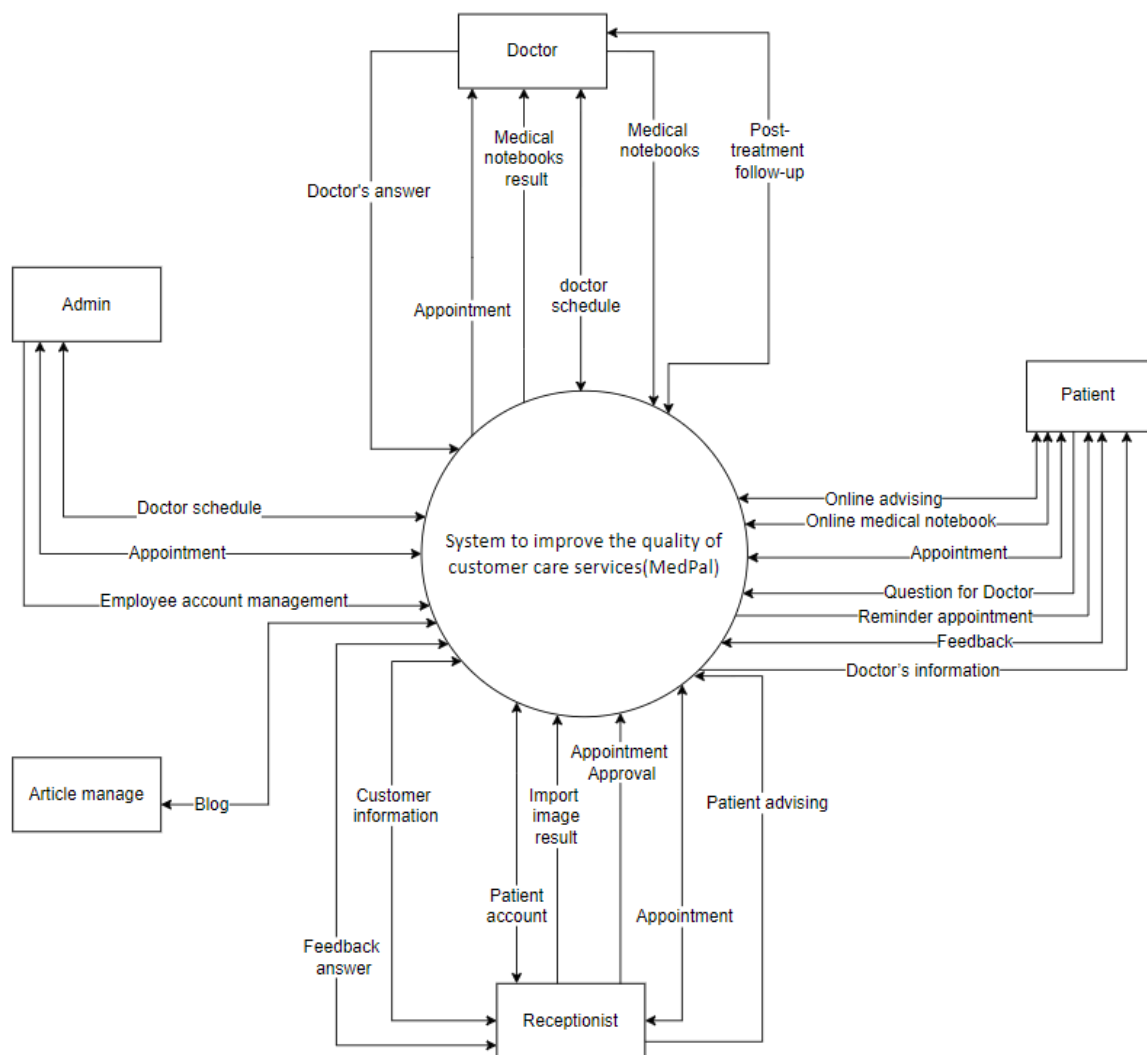


Figure 1.1.1 MedPal Context diagram

1.2 Main Workflows

1.2.1 Appointment processing

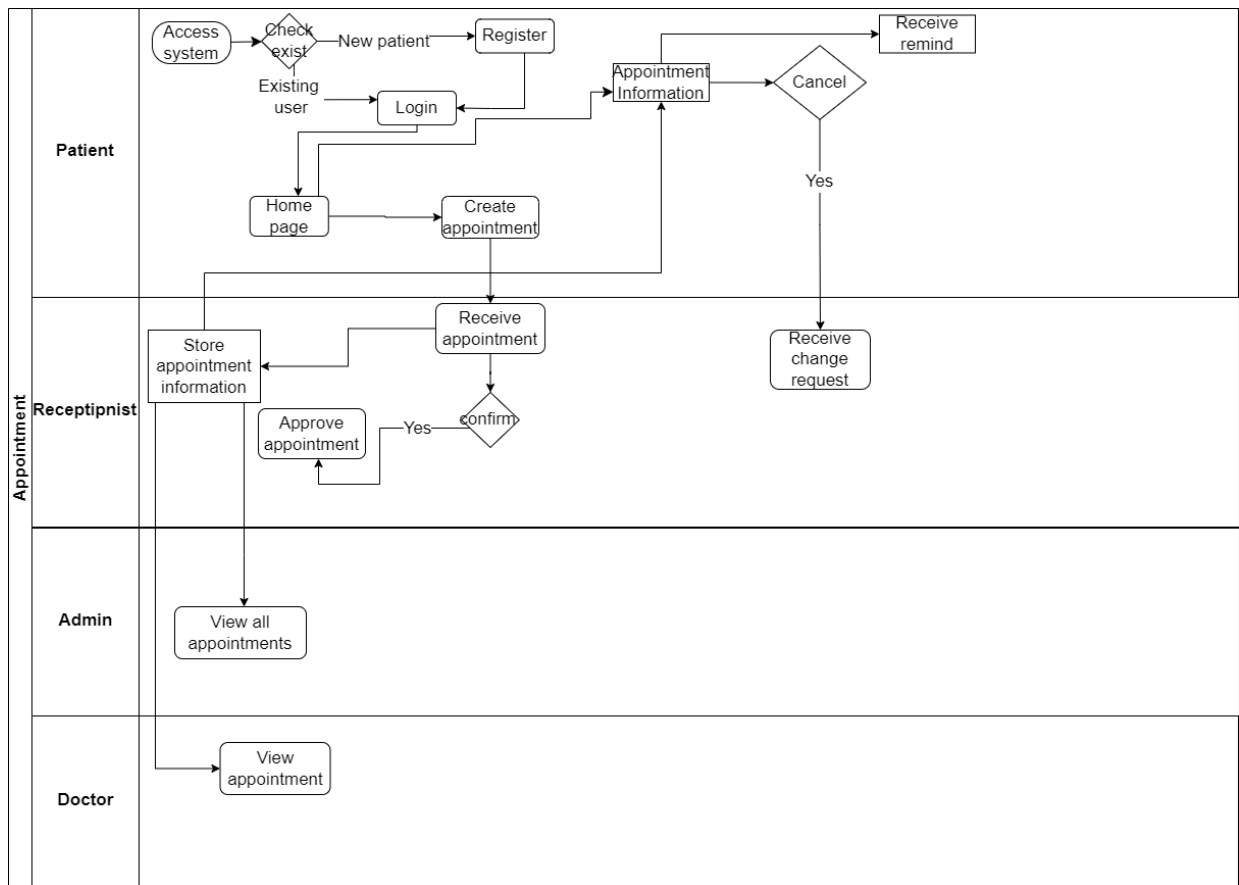


Figure 1.2.1.1 Appointment processing workflow

1.2.2 Online advising

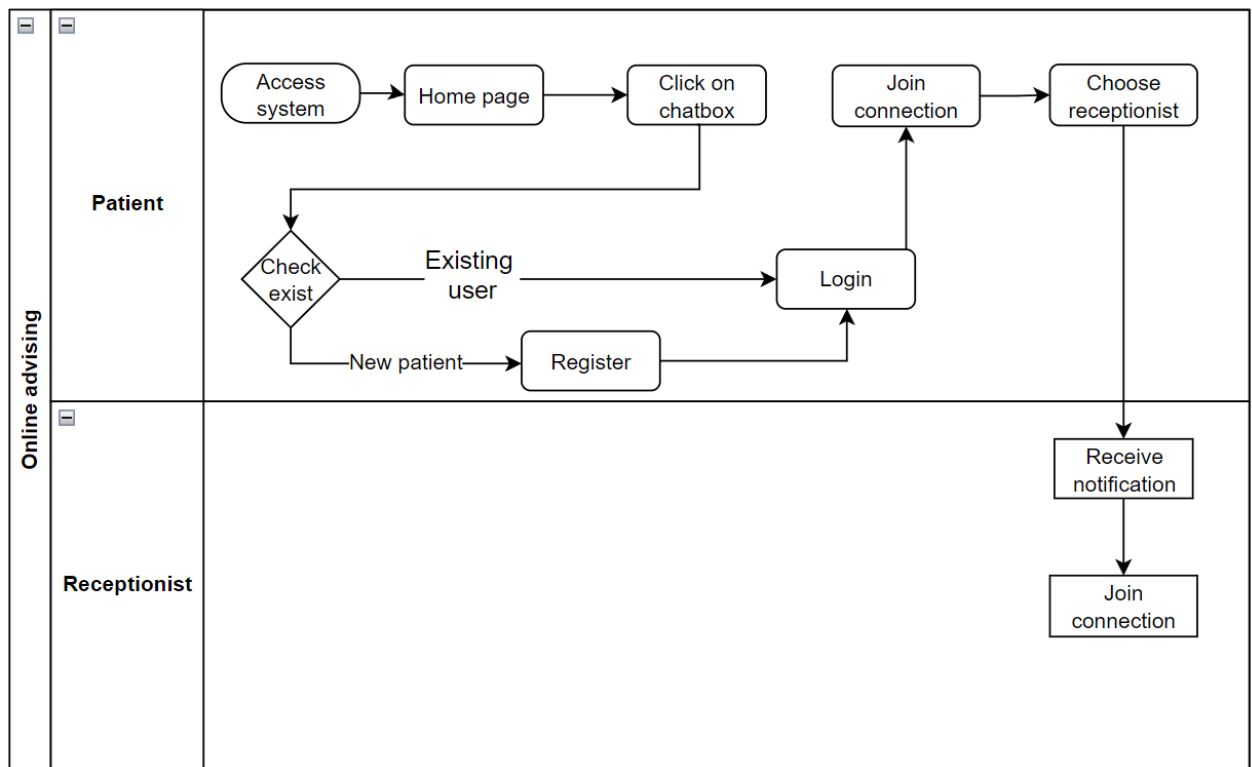


Figure 1.2.2.2 Online advising workflow

1.2.3 Store medical notebooks

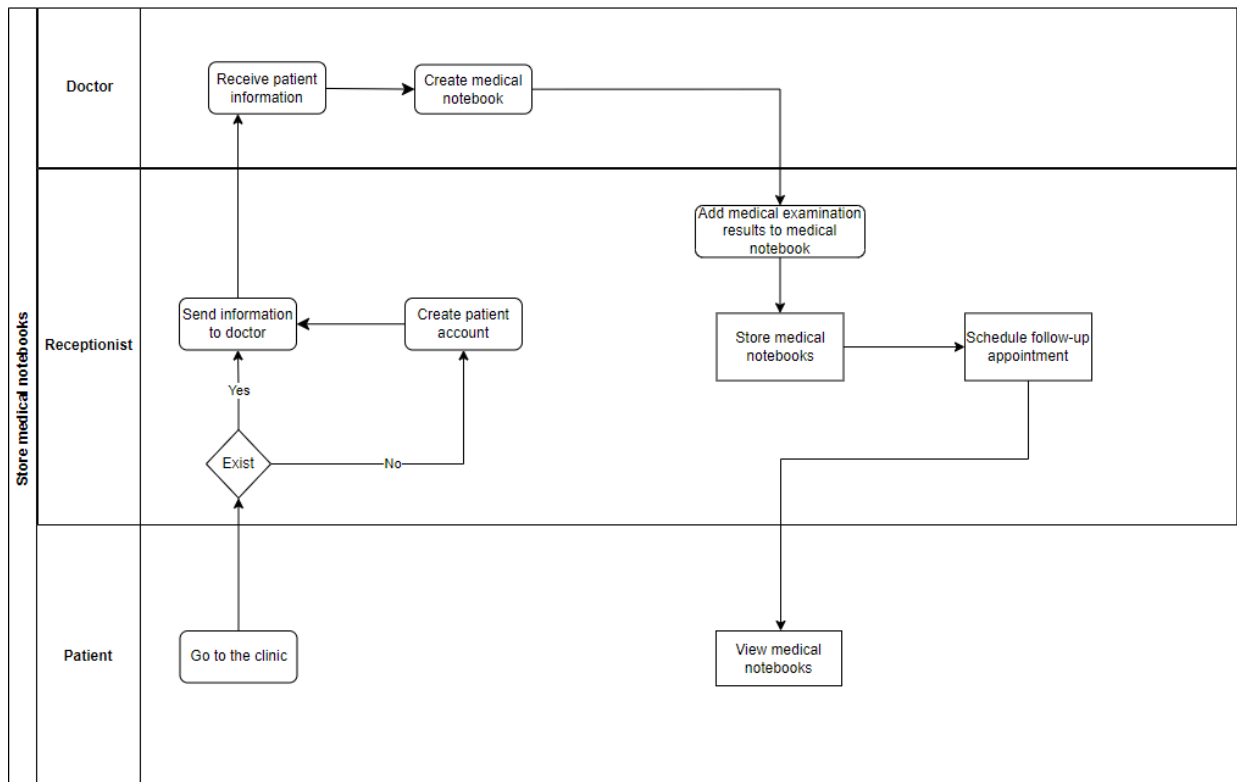


Figure 1.2.3.1 Store medical notebooks workflow

1.2.4 Doctor schedule

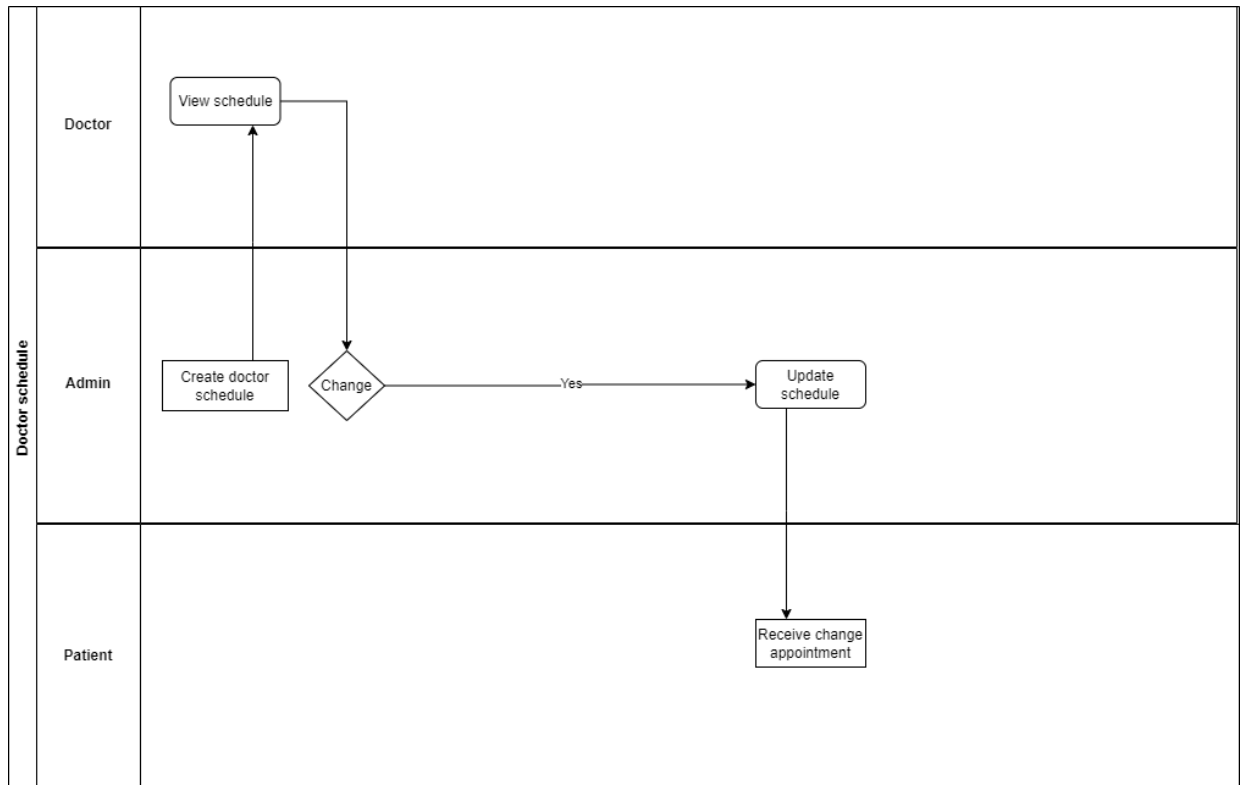


Figure 1.2.4.1 Doctor schedule workflow

1.2.5 Ask question

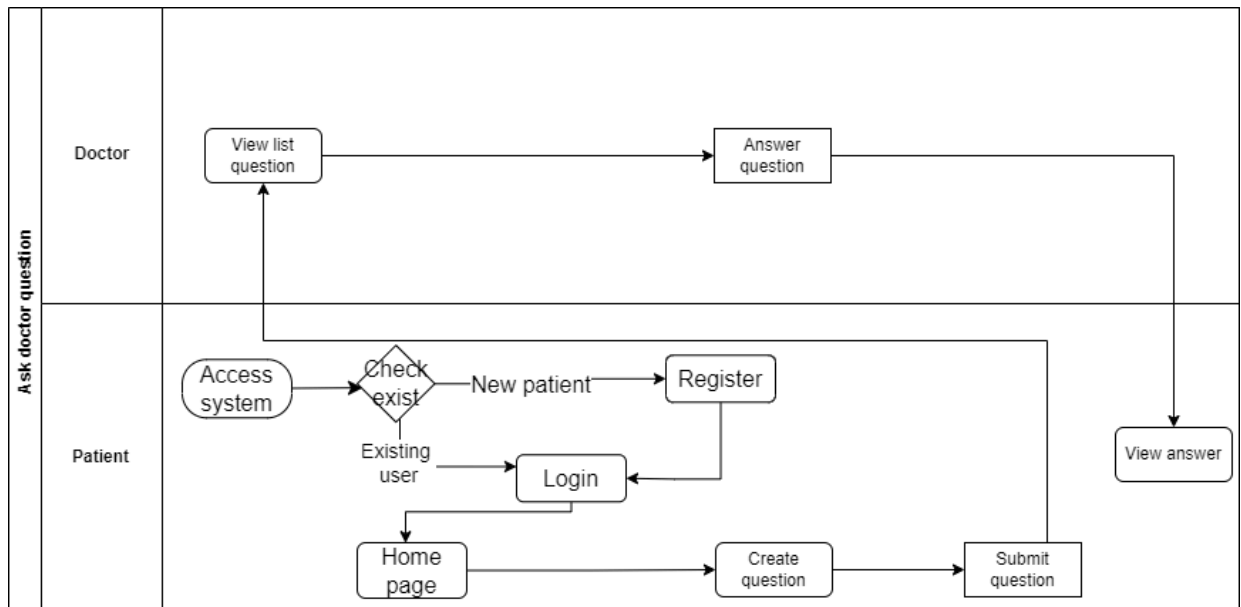


Figure 1.2.5.1 Ask question workflow

1.2.6 Post treatment follow up

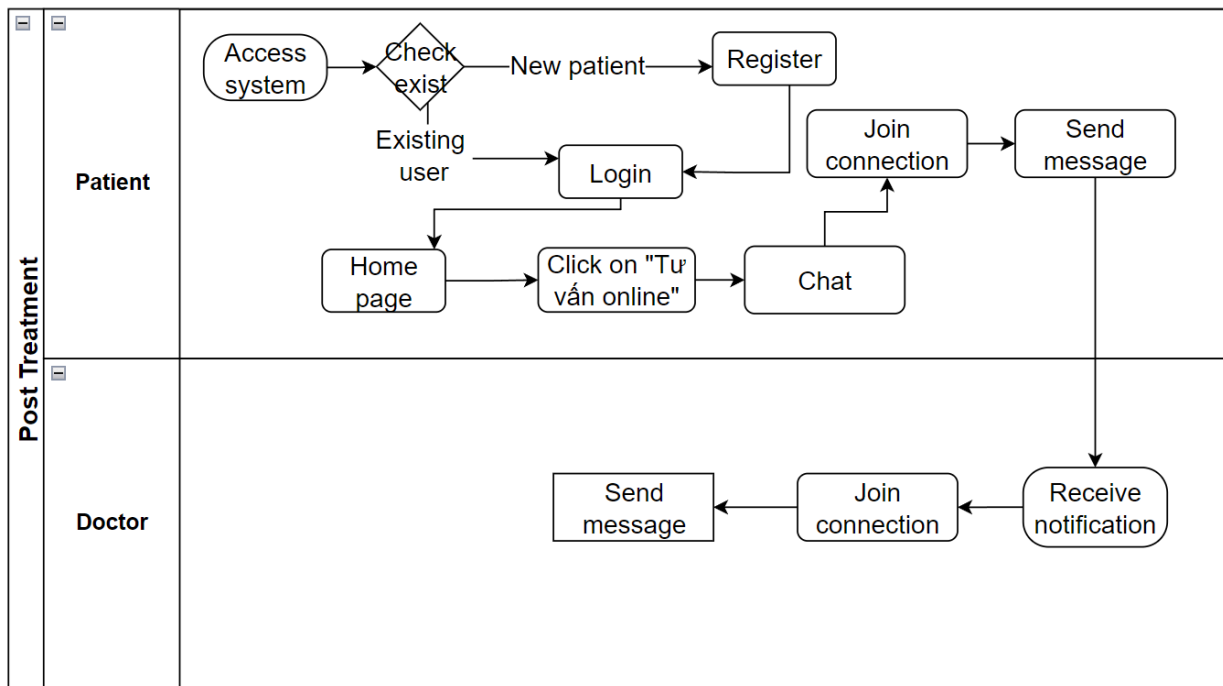


Figure 1.2.6.1 Post treatment follow up workflow

1.3 User Requirements

1.3.1 Actors

Table 1.3.1.1 describe a list of actors that interact with IT service management system, it includes Administrators, Doctors, Patients, Receptionists, and Article manager applications

Table 1.3.1.1: MedPal Distributed Management Actor List

#	Actor	Description
1	Administrators	The person with the highest authority. Besides, he can manage doctors schedules and online appointments ,view feedback and quality control.
2	Doctors	The user can choose their own schedule and appointments, and manage medical results and blog information.
3	Patients	They choose a schedule to make an appointment. In addition,they are able to view information like medical results, records, feedback and create feedback.
4	Receptionists	The user primarily feedback and reminder appointments for the doctors.
5	Article manager	who writes blogs, and writes articles introducing medical examination and treatment processes

1.3.2 Use Cases

1.3.2.1 Administrators

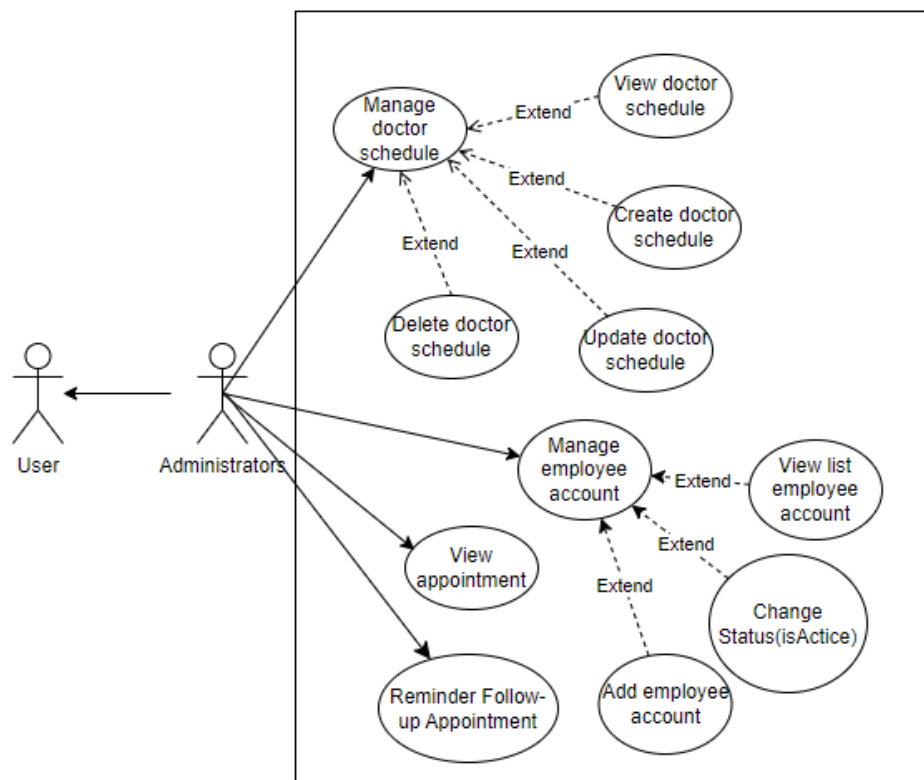


Figure 1.3.2.1.1 Administrators use case diagram

1.3.2.2 Doctor

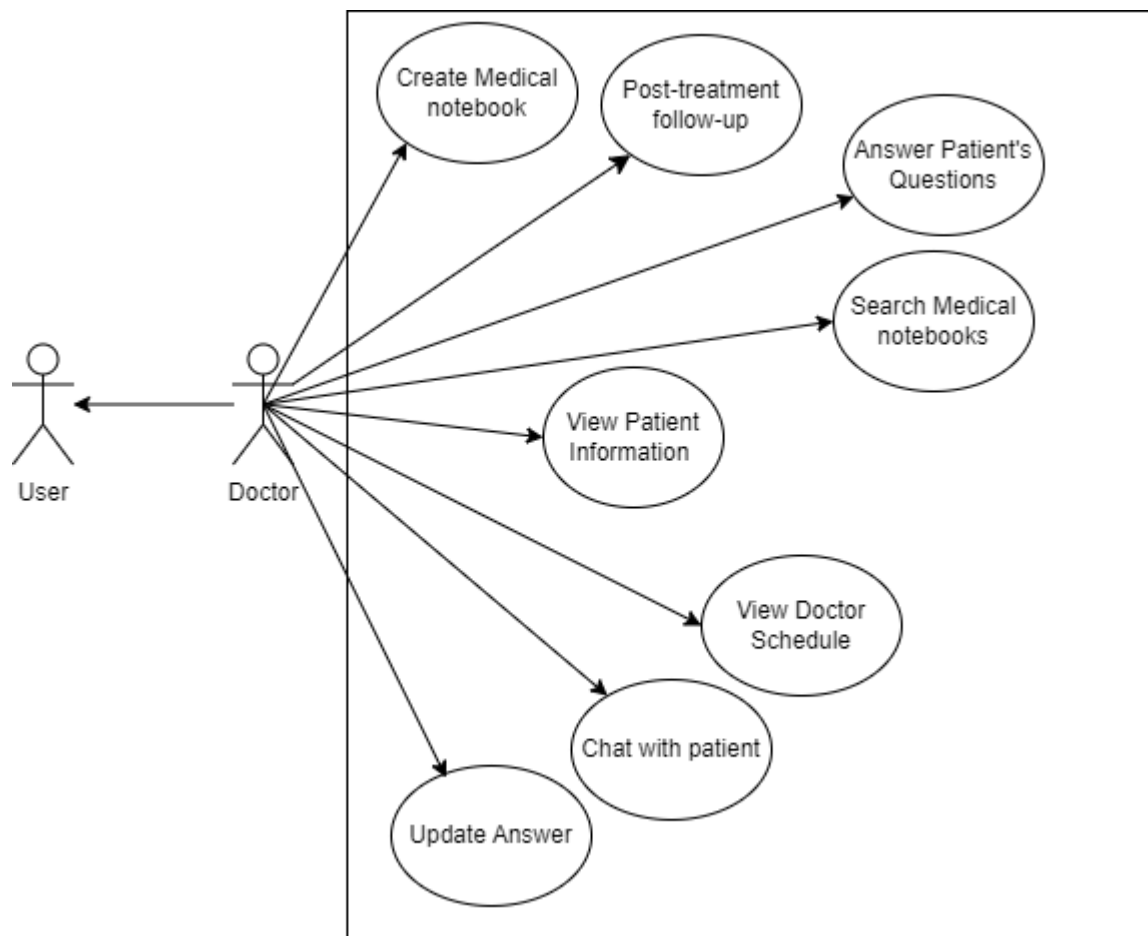


Figure 1.3.2.2.1 Doctor use case diagram

1.3.2.3 Patient

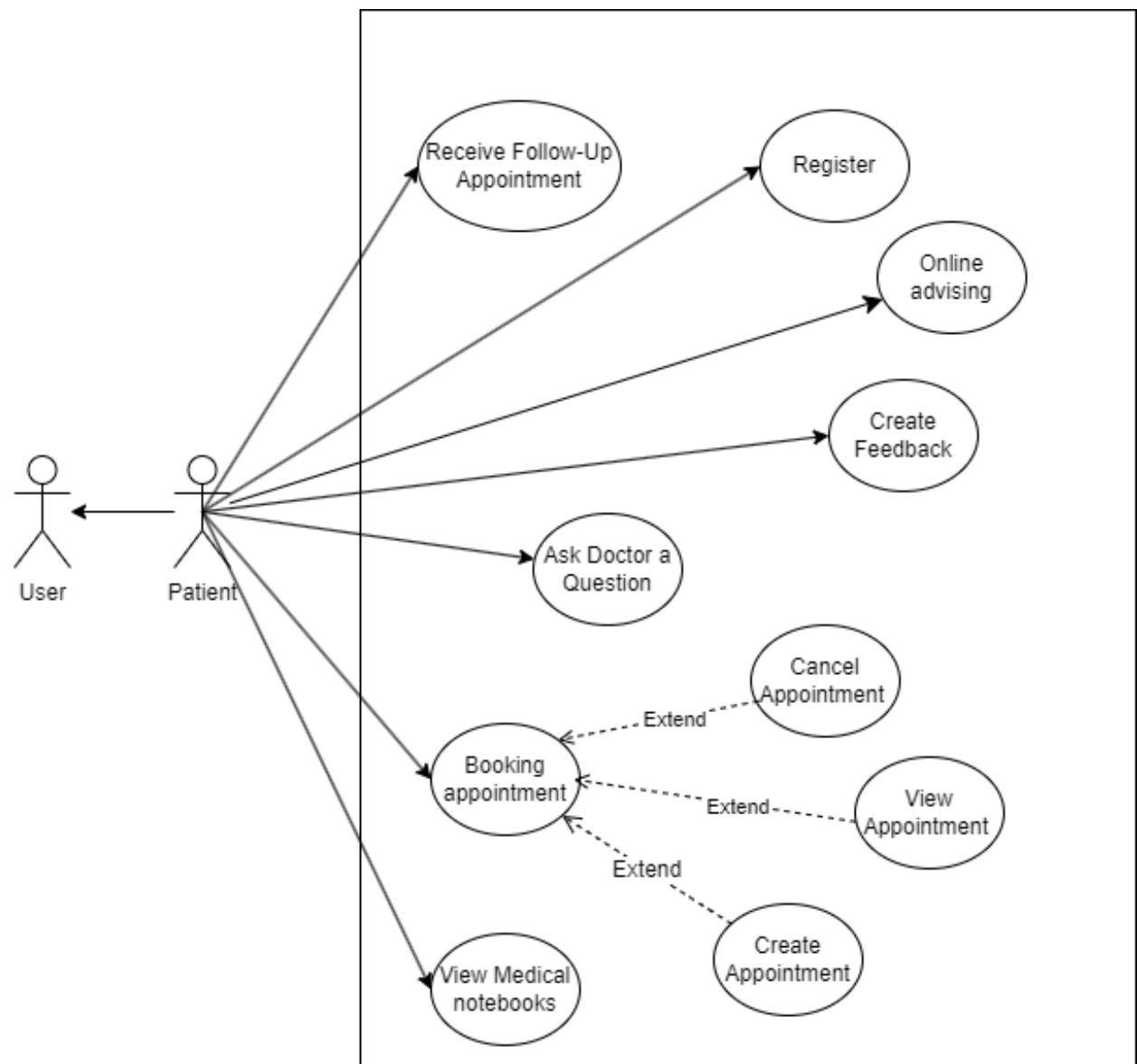


Figure 1.3.2.3.1 Patient use case diagram

1.3.2.4 Article management

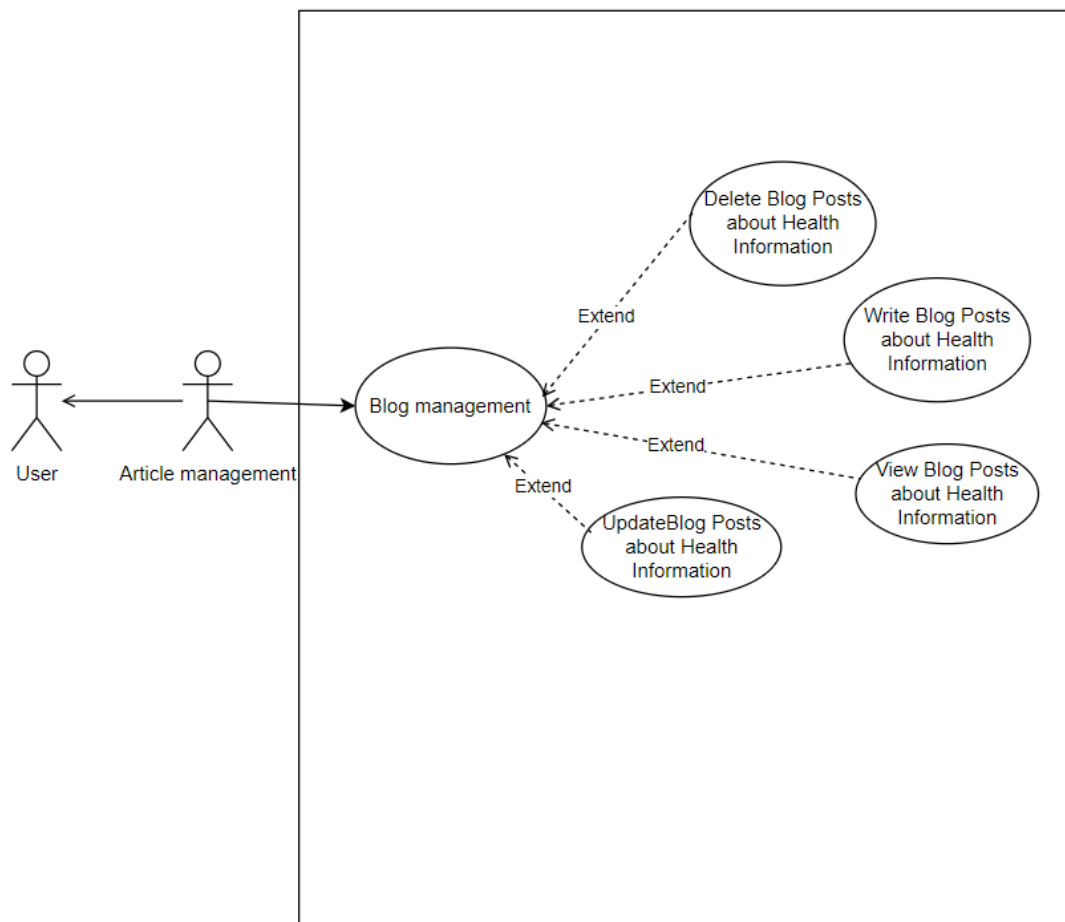


Figure 1.3.2.4.1 Article manager use case diagram

1.3.2.5 Receptionist

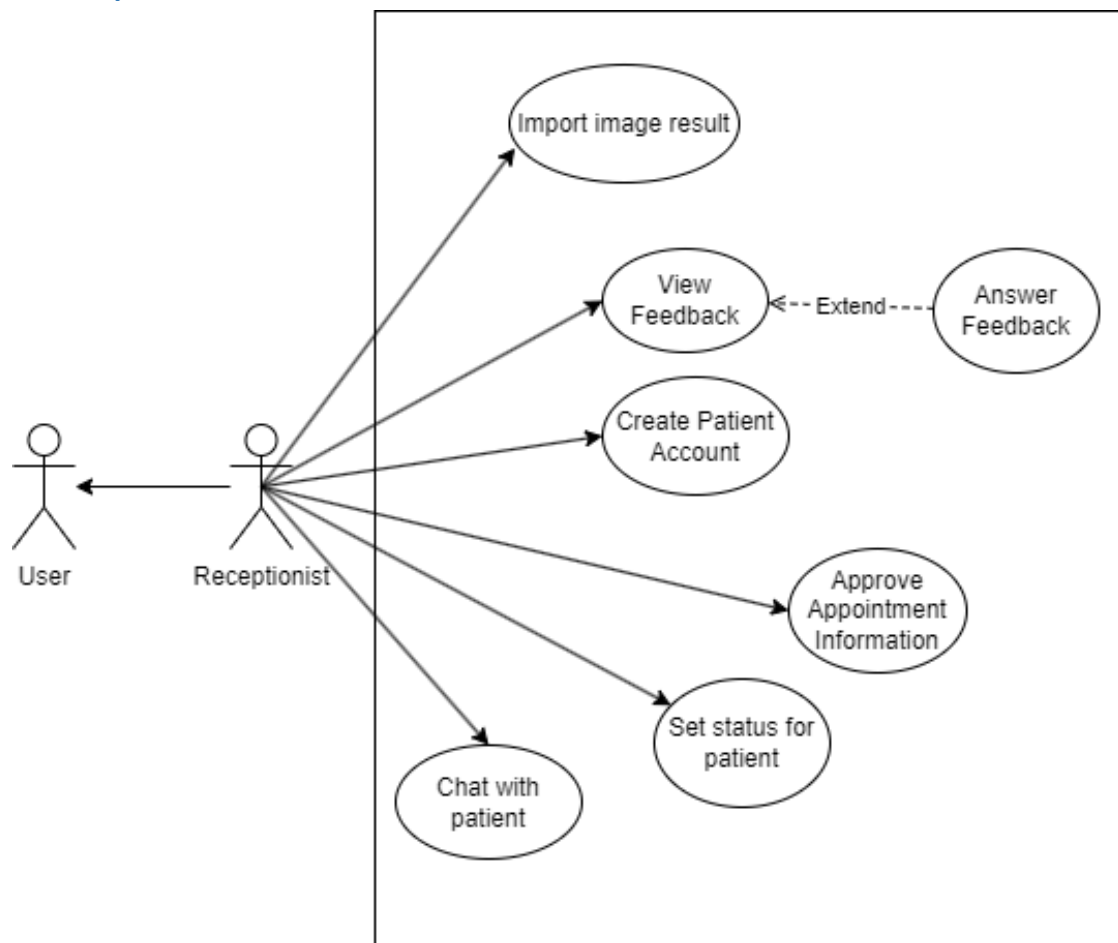


Figure 1.3.2.5.1 Receptionist use case diagram

1.3.2.6 User

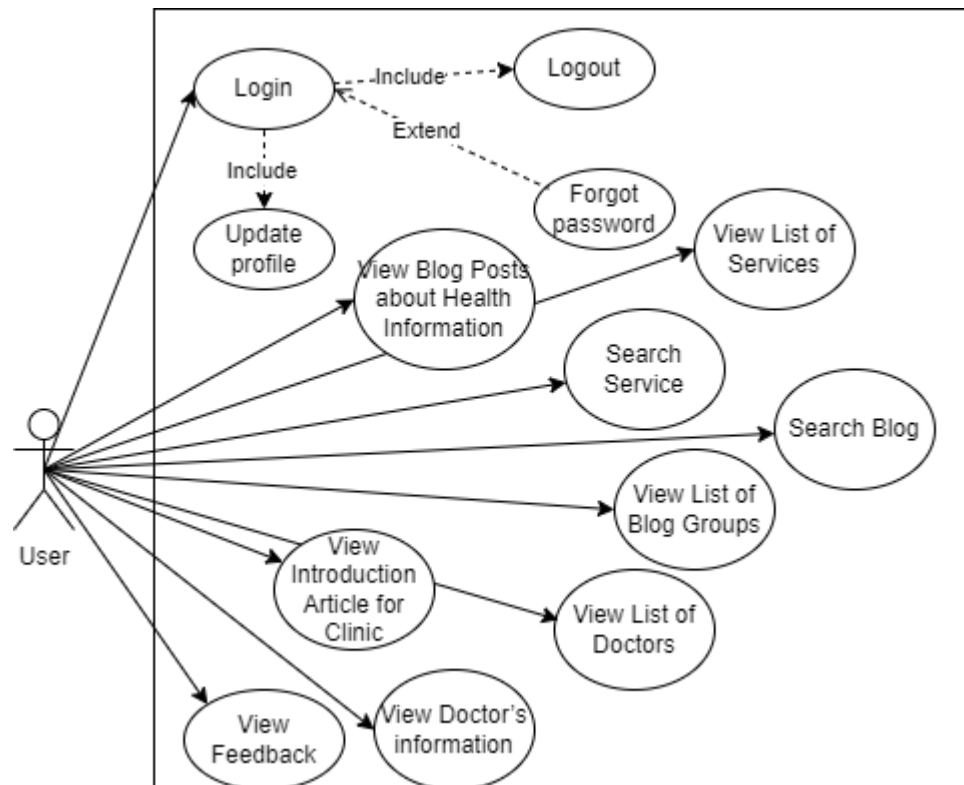


Figure 1.3.2.6.1 User use case diagram

1.4 System Functionalities

1.4.1 Patient Screens Flow

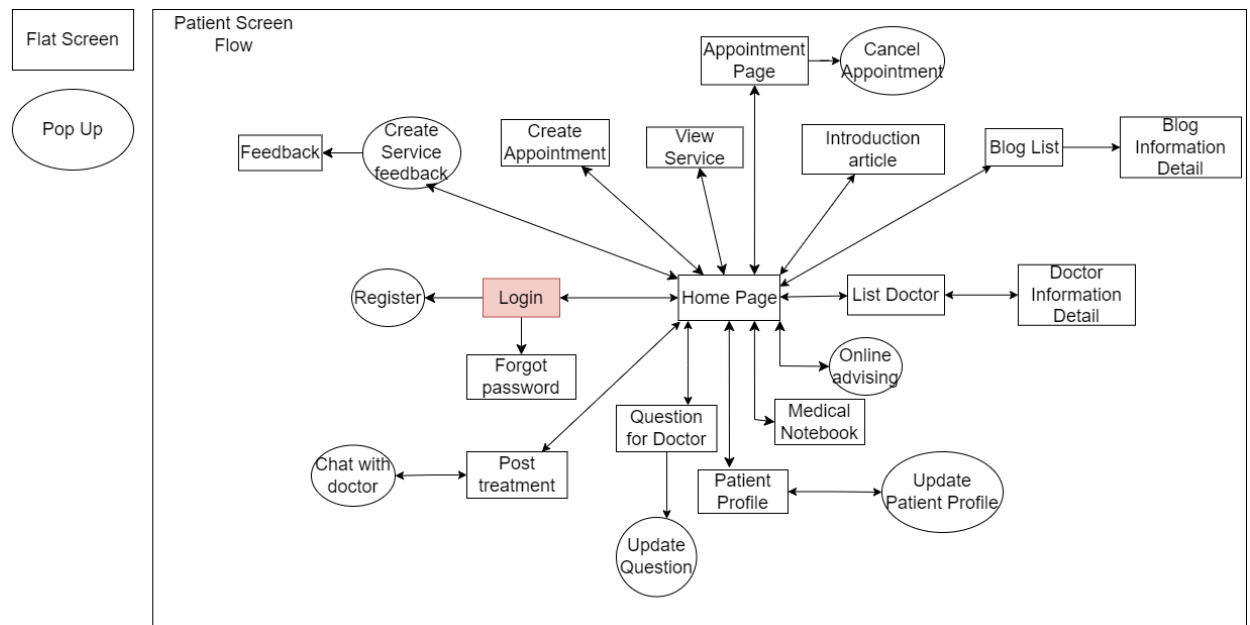


Figure 1.4.1.1 Patient screen flow

1.4.2 Doctor Screens Flow

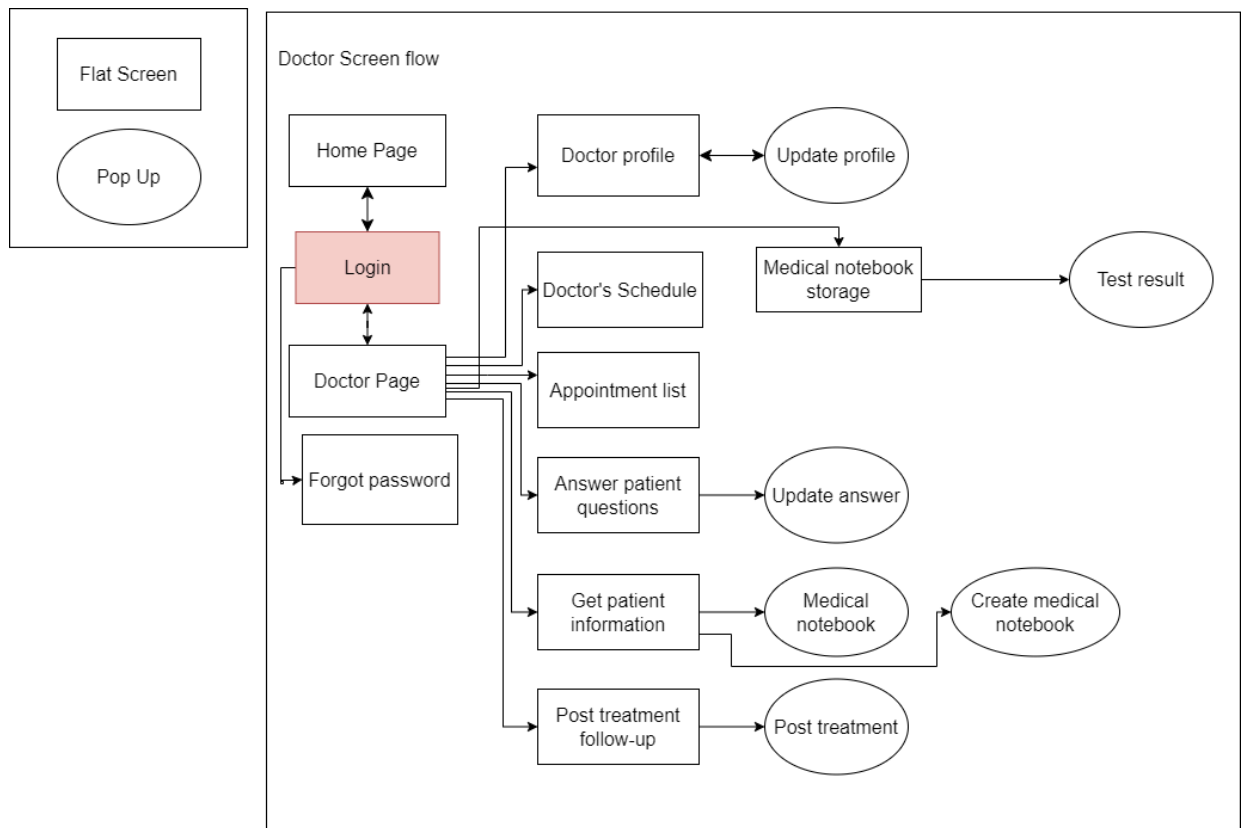


Figure 1.4.2.1 Doctor screen flow

1.4.3 Article manager Screens Flow

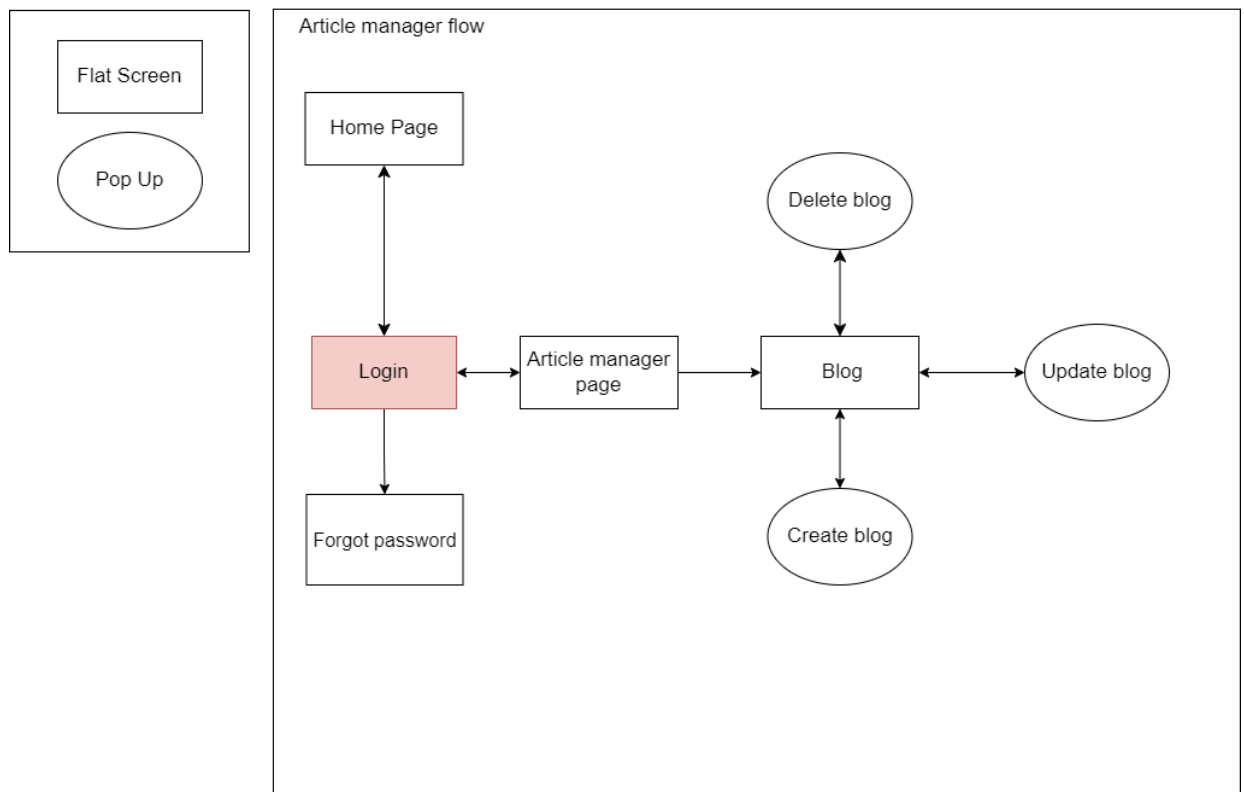


Figure 1.4.3.1 Article manager screen flow

1.4.4 Receptionist Screens Flow

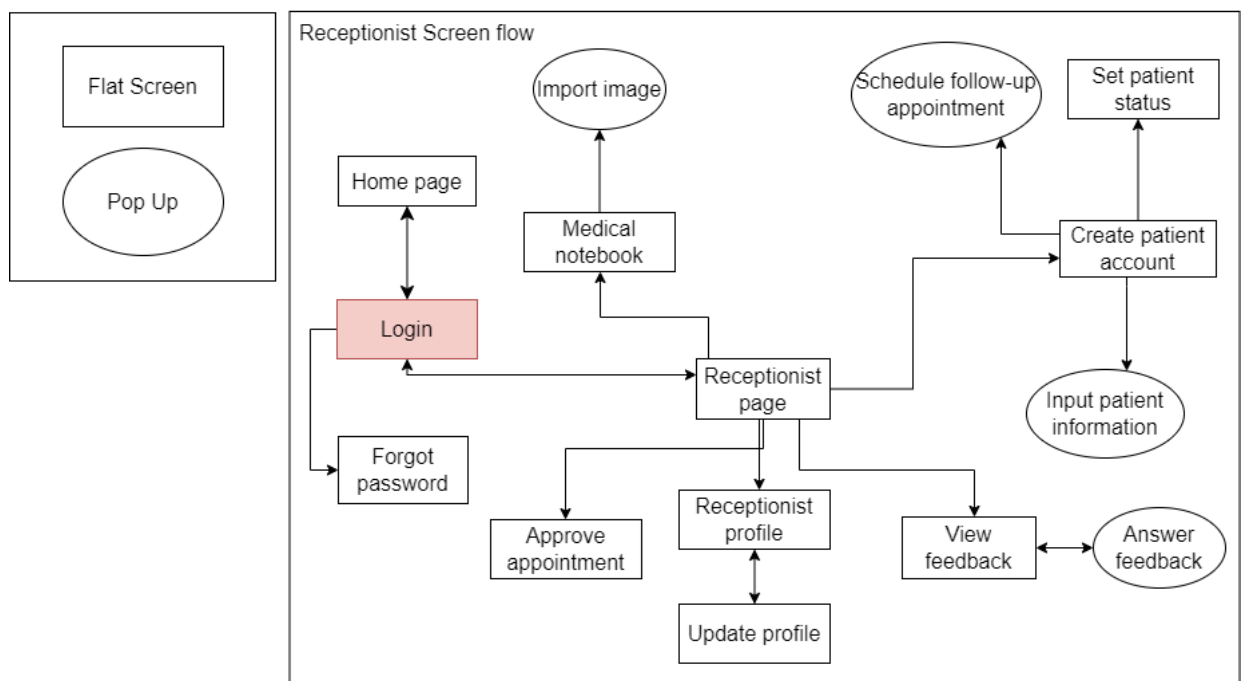


Figure 1.4.4.1 Patient screen flow

1.4.5 Admin Screens Flow

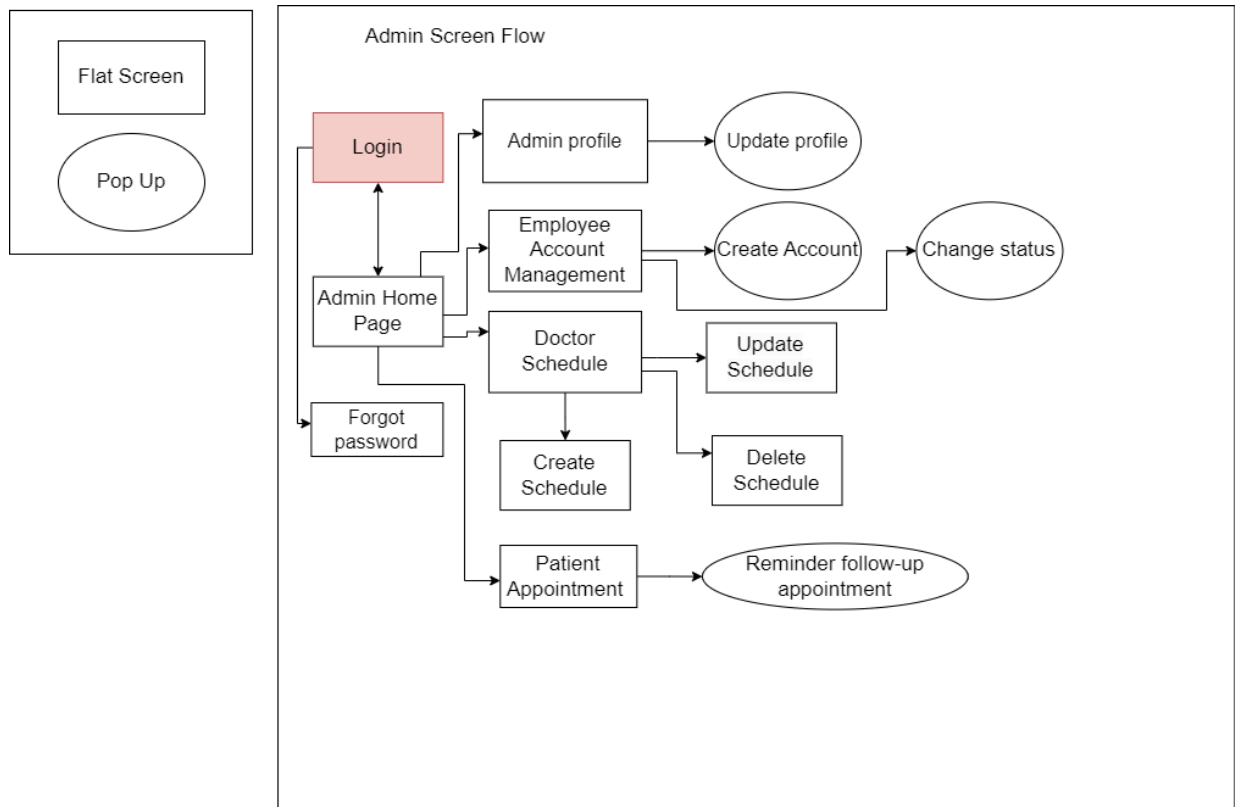


Figure 1.4.5.1 Admin screen flow

1.4.6 Screen Authorization

Tabel 1.4.6.1 Screen Authorization

Screen	Admin	Patient	Receptionist	Doctor	Article manager
Home Page	X	X	X	X	X
List doctor		X			
Doctor information		X			
Login	X	X	X	X	X
Register		X			
Forgot password	X	X	X	X	X
Online advising		X			
Medical notebook		X			
Patient profile		X			
Question for doctor		X			
Post treatment		X			
Chat with doctor		X			
Update question		X			
Update patient profile		X			
Create service feedback		X			

Feedback		X			
View service		X			
Create appointment		X			
Introduction article		X			
Blog list		X			
Blog information		X			
Appointment page		X			
Cancel appointment		X			
Doctor page				X	
Doctor profile				X	
Update profile				X	
Doctor's schedule				X	
Medical notebook storage				X	
Test result				X	
Appointment list				X	
Answer patient questions				X	
Update answer				X	
Get patient information				X	
Medical notebook				X	
Create medical notebook				X	
Post treatment follow-up				X	
Post treatment				X	
Admin home page	X				
Admin profile	X				
Update profile	X				
Employee account management	X				
Create account	X				
Change status	X				
Doctor schedule	X				
Create doctor schedule	X				
Update doctor schedule	X				
Delete doctor schedule	X				
Patient appointment	X				
Reminder follow-up appointment	X				
Article manager home page					X
Blog					X
Delete blog					X
Update blog					X
Create blog					X
Receptionist page			X		
Approve appointment			X		
Receptionist profile			X		
Update profile			X		

View feedback			X		
Answer feedback			X		
Create patient account			X		
Input patient information			X		
Set patient status			X		
Schedule follow-up appointment			X		
Medical notebook			X		
Import image			X		

1.4.7 Non-UI Functions

Table 1.4.7.1 Non-UI Functions

#	Feature	System Function	Description
1	Notification	SMS notification service	Send to a selected phone number for reminding appointments, medical results.
2	Authentication	Authentication service	Manages user authentication and authorization, ensuring secure access to the system.
3	Security	Encryption service	Encrypt sensitive data such as user password, and personal information for secure storage and transmission
4	Error handling	Error logger	Logs errors and exceptions, triggering alerts for critical issues and providing information for troubleshooting.
5	API Integration	External API Handler	Manages interactions with external APIs for services such as third-party integrations.
6	Data backup and recovery	Recovery service	Provides backup and recovery solutions to protect against data loss and ensure system resilience.
7	Connecting between GUI and server	API Connection	Connect data between GUI and back-end server

1.5 Entity Relationship Diagram

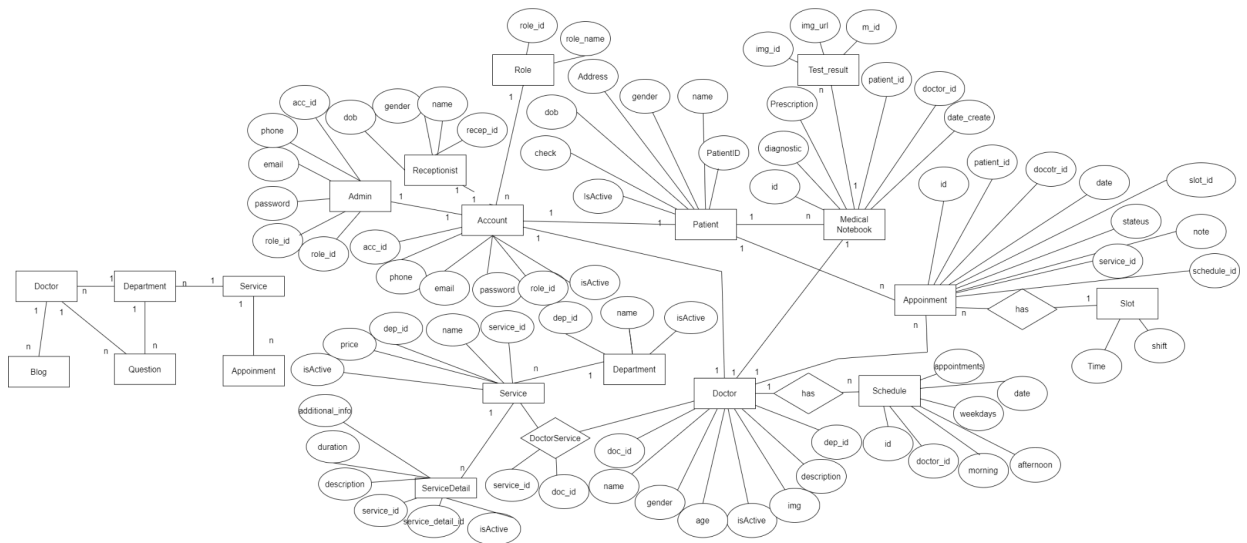


Figure 1.5.1 Entity Relationship Diagram

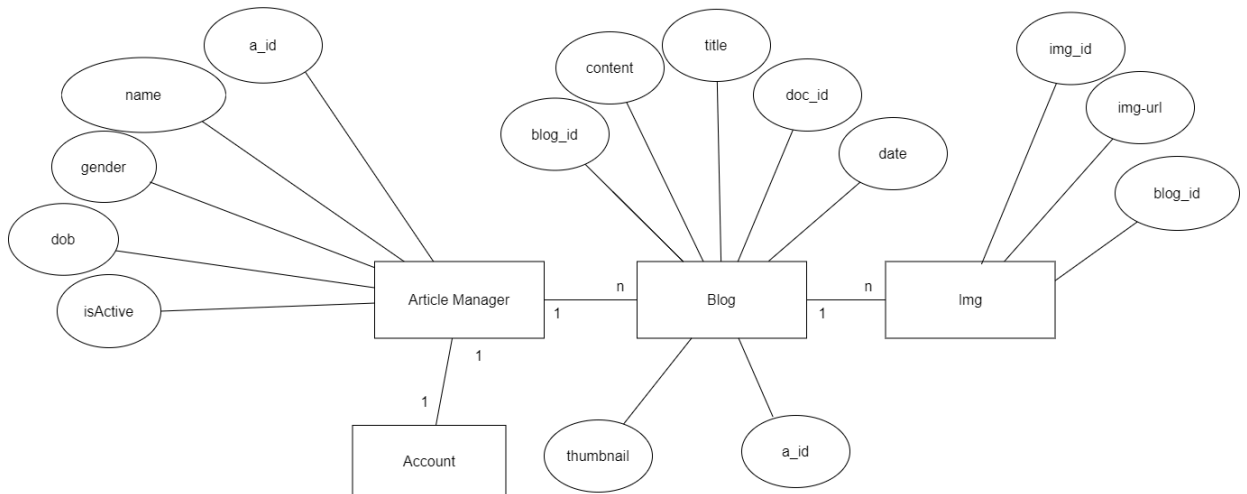


Figure 1.5.2 Entity Relationship Diagram for Article Manager

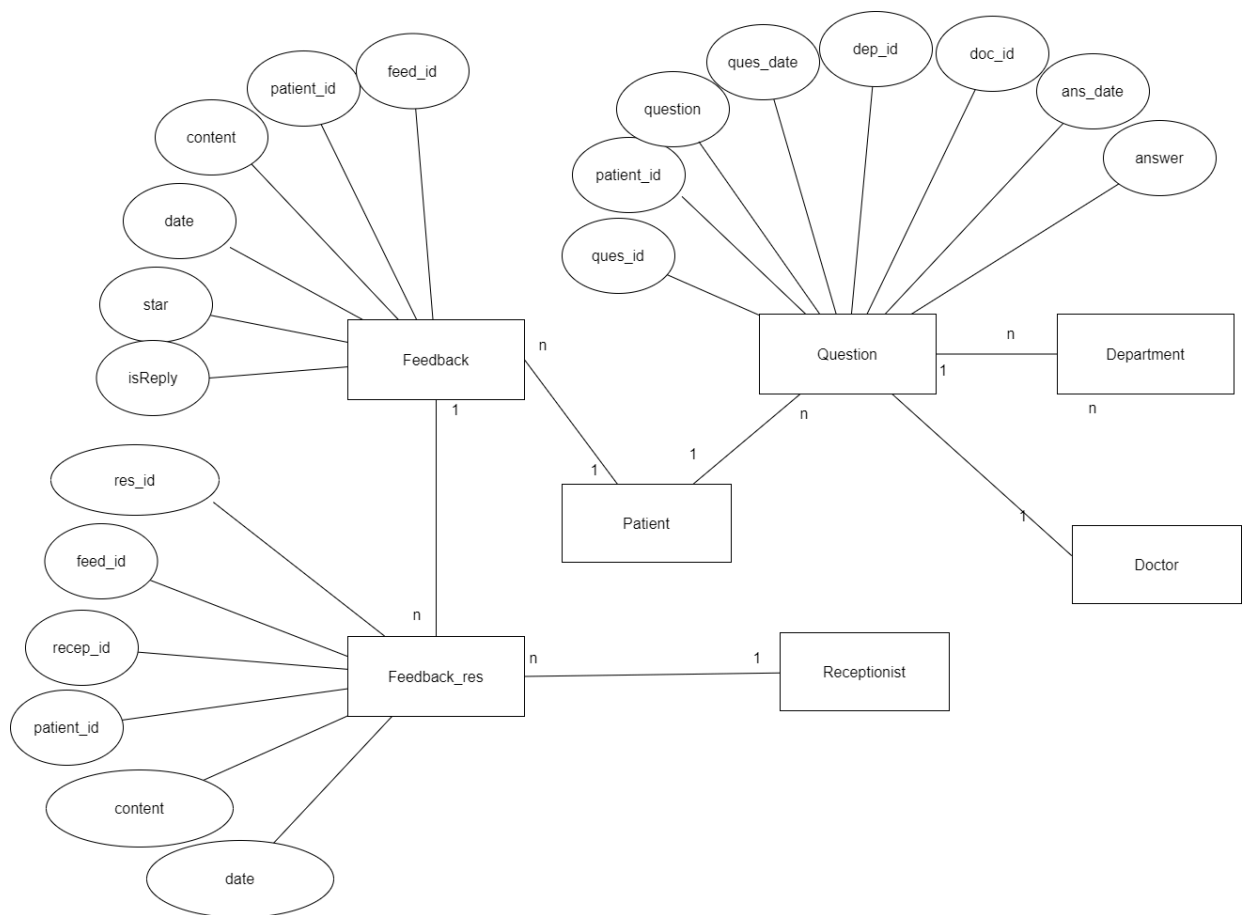


Figure 1.5.3 Entity Relationship Diagram for Question & Feedback



Figure 1.5.4 Entity Relationship Diagram for Message

2. Use Case Specification

2.2 Common Feature

2.2.1 Login System

Table 2.2.1.1 Login system use case

Primary Actors	All users	Secondary Actors	None
Description	As a user, I want to be able to log into the system so that I can use the system's authenticated features and access my personalized account.		
Preconditions	User account has been created & authorized		

Postconditions	• User logs in the system successfully • The system tracked successful login into the Activity Log
Normal Flows	Login System 1. User clicks "Đăng nhập" button from the page header or accesses an authenticated feature (from a link or type the page URL directly into the address bar) 2. User accesses the User Login screen 3. User types in the login details or choo other login options 4. User clicks the "Đăng nhập" button 5. System validates the login details 6. System allows user to access 7. System tracks user's success login to the Activity Log 8. System accesses the Home Page (or the previous calling page if any)
Alternative Flows:	Step 2.1_ Google Login 1. User clicks Google "Đăng nhập" button to login system using Google account 2. System redirects the user to the Google's Login screen 3. User types in the Google account details and chooses to login 4. Google validates user's login information successfully and redirect him/her back to the system 5. Return to step 5 of normal flow. Step 4_System can't authenticate the user User can't be authenticated & get relevant error message in one of below cases 1. He/she leaves the Email and/or Password field blank (MSG10) 2. Input Email or Password are incorrect (MSG09) 3. Input Email/Password are correct but email has not been verified (MSG11) 4. The user account is blocked / inactive (MSG12) If user inputs wrong logging-in details 5 times continuously, system will lock his/her account in 30 minutes (with relevant warning message - MSG13)

2.2.2 Log Out

Table 2.2.2.1 Log Out use case

Primary actor:	All User	Secondary actor:	None
Description:	This use case describes the process and associated interactions when a user initiates the logout action in an MedPal system		
Trigger:	1. The User click button “Đăng xuất” in nav bar		
Pre-condition:	1. The user is currently logged in to the system		
Post-condition:	1. The current session will be deleted and redirected to the login page		
Normal flow:	1. The system terminates the user’s session. 2. The system redirects to Login Page		
Alternative flows:	None		

2.2.3 Forgot Password

Table 2.2.3.1 Forgot Password use case

Primary actor:	All User	Secondary actor:	None
Description:	The purpose of this use case is to allow User to set a new password for an account in case they lose their password.		
Trigger:	1. User clicks on the "Quên mật khẩu" link on the login or password recovery page.		
Pre-condition:	1. The user must have an account registered in the system		
Post-condition:	1. The user's password is successfully reset, and they can log in with the new password.		
Normal flow:	1. The system shows the reset password form including the new password and reenters the new password. 2. The user enters a new password and reenters the new password. 3. The system validates the new password and updates it for the user's account. 4. The system confirms the successful password reset.		

Alternative flows:	1. Invalid password (Passwords must have at least 8 characters, including at least one number and one special character.): Systems notify: “Passwords must have at least 8 characters, including at least one number and one special character. please re-enter the password”. 2. Users enter the re-enter password isn’t the same as the password: Systems notify: “The re-enter password is not the same as the password. Please re-enter the password”.
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2.2.4 Update Profile

Table 2.2.4.1 Update profile use case

Primary actor:	All User	Secondary actor:	None
Description:	This use case allows users to manage information about their accounts such as passwords, and personal information,...		
Trigger:	1. The user click on “Tài khoản” icon in navbar		
Pre-condition:	1. The user is currently logged in to the system		
Post-condition:	None		
Normal flow:	1. User click on account logo on the right of navbar, after user choose “Cập nhật thông tin”, the system will lead user to the new page 2. They will see their information, fill in all fields that user want to change, and click “Lưu” to save change		
Alternative flows:	1. If the user edits required fields to be blank, it will not be possible to update and save changes.		

2.2.5 View Patient Information

Table 2.2.5.1 View Patient Information use case

Primary actor:	Doctor	Secondary actor:	None
Description:	This use case describes the process of Doctors can view patient information about entering the examination through the display screen, including the patient's personal information and Medical Notebooks.		
Trigger:	1. The doctor clicks on “Xem”		
Pre-condition:	1. The doctor has logged into the system		

Post-condition:	None
Normal flow:	1. The doctor selects "Xem bệnh nhân đang hoạt động" in sidebar and click on "Xem". 2. The system displays all medical notebooks for that patient.
Alternative flows:	1. Doctors click on "Tương tác với các bệnh nhân đã khám" and select "Xem hồ sơ bệnh án" 2. The system displays all medical notebooks for that patient.

2.2.6 Chat with patient

Table 2.2.6.1 Chat with patient use case

Primary actor:	Receptionist	Secondary actor:	None
Description:	This use case describes the process of a doctor prescribing medication to a patient and recording a diagnosis		
Trigger:	1. The doctor clicks on the "Thêm mới"		
Pre-condition:	1. The doctor must be logged in to the system		
Post-condition:	The notification page will be shown over the current page.		
Normal flow:	1. The doctor clicks on the "Thêm mới", which will have basic fields such as the "Toa thuốc", and "Chẩn đoán", and the doctor need to fill full of 2 fields.. 2. Once completed, the doctor can click "Tạo" to send the attached prescription to the patient		
Alternative flows:	1. If the doctor does not fill in any fields or only 1 of 2, the system will not allow the creation of a diagnosis		

2.2.7 Search Medical Notebooks

Table 2.2.7.1 Search Medical Notebooks use case

Primary actor:	Doctor	Secondary actor:	None
Description:	This use case describes the process of searching for Medical Notebook information		
Trigger:	1. Users will search Medical Notebooks on the Trang chủ > Tìm kiếm kết quả		

Pre-condition:	1. The User is currently logged in to the system
Post-condition:	
Normal flow:	1. When clicking "Tra cứu kết quả", the user will be taken to a screen with 3 information fields: "Ngày khám, Mã tra cứu, Mã bảo mật", enter what the user wants to search for here 2. After pressing "Tìm kiếm", if there are results, the system will redirect the user to the page with the corresponding Medical Notebook
Alternative flows:	1. If no results are found, the system returns the message "Không thể tìm thấy kết quả tương tự"

2.2.8 View Medical Notebooks

Table 2.2.8.1 View Medical Notebooks use case

Primary actor:	Patient	Secondary actor:	None
Description:	Allows view the Medical Notebooks of patients		
Trigger:	1. The User click button “Tra cứu kết quả bệnh án”		
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. The User click button “Xem Sổ tay y tế” 2. The system displays all of the Medical Notebook of the patient, showing patient information such as patient name, ordering physician, indication, diagnosis		
Alternative flows:	If no results are found, the system returns the message "Không thể tìm thấy"		

2.2.9 View Doctor Schedule

Table 2.2.9.1 View Doctor Schedule use case

Primary actor:	Admin	Secondary actor:	None
Description:	Allows view all the doctors schedule in a week		
Trigger:	1. The User clicks button “Xem lịch làm việc ”		
Pre-condition:	1. The user is logged into the system and has the corresponding		

	permissions.
Post-condition:	None
Normal flow:	1. The User clicks button “Xem lịch làm việc” 2. The user choose start date and end date 3. Click on “GỬI” to view schedule that match with start date and end date 4. The system displays the Schedule of doctors, showing information of schedule such as time, slot, ID..
Alternative flows:	None

2.2.10 View Appointment

Table 2.2.10.1 View Appointment use case

Primary actor:	Admin	Secondary actor:	None
Description:	Allows doctors to view the appointments of patients		
Trigger:	The User clicks button “Xem lịch khám”		
Pre-condition:	The user is logged into the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. The User clicks button “Xem lịch khám” 2. The system displays the appointment schedule of the patient, showing information of schedule such as time, slot, ID, patient information...		
Alternative flows:	None		

2.2.11 Answer Patient Questions

Table 2.2.11.1 Answer Patient Questions use case

Primary actor:	Doctor	Secondary actor:	None
Description:	Allows the doctor to answer the questions of the guest		
Trigger:	1. The User clicks to the “Trả lời câu hỏi bệnh nhân” and choose pen icon to write the comment to answer the question		
Pre-condition:	1. The user is logged into the system and has the corresponding permissions.		

Post-condition:	None
Normal flow:	1. The User clicks to the "Trả lời câu hỏi bệnh nhân" and choose pen icon to write the comment to answer the question. 2. After the answer is already write down, the doctor click on "Lưu" to send the answer for guest question
Alternative flows:	1. The doctor click on "Hủy" 2. Everything written will be cancelled by the system

2.2.12 Create Doctor Schedule

Table 2.2.12.1 Create Doctor Schedule use case

Primary actor:	Administrators	Secondary actor:	None
Description:	Allow the admin to create a Schedule for doctor		
Trigger:	1. the user clicks to "Lưu lịch làm việc" button		
Pre-condition:	The user is logged into the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. User clicks the "Lưu lịch làm việc" button then the system will redirect the user to create a page to enter a new Schedule. 2. User enter Schedule information and then click the "Lưu lịch làm việc" button to create a new Schedule. 3. The system validates the Schedule input information. 4. System added a new Schedule . 5. System notify users of successful action.		
Alternative flonews:			

2.2.13 Update Doctor Schedule

Table 2.2.13.1 Update Doctor Schedule use case

Primary actor:	Administrators	Secondary actor:	None
Description:	Allow the admin to update the Schedule for doctor		
Trigger:	1. the user clicks to "Chỉnh sửa lịch làm việc " button		

Pre-condition:	The user is logged into the system and has the corresponding permissions.
Post-condition:	None
Normal flow:	1. User clicks the “Chỉnh sửa lịch làm việc” button then the system will show the user to update pop-up to enter new information. 2. User enter new Schedule information and then click the “Chỉnh sửa” button to update a Schedule. 3. The system validates the Schedule input information. 4. System update a Schedule . 5. System notify users of successful action.
Alternative flows:	None

2.2.14 Change Status of Doctor(isActive)

Table 2.2.14.1 Change Status of Doctor(isActive) use case

Primary actor:	Administrators	Secondary actor:	None
Description:	Allow the admin to change doctor status, if isActive = true, that mean this account has permission and active		
Trigger:	1. User click icon “Chỉnh sửa” on the side of account information row		
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. The system show list of Doctor and action change beside the doctor information 2. The admin choose “Active” or “Inactive” 3. The user clicks “Chỉnh sửa” and “Lưu” on the popup show to confirm the change schedule. 4. The system updates the status of the doctor. 5. The system notified users of successful action.		
Alternative flows:	2.1. Users click “Cancel” on the popup notification. 2.2 The system cancelled the delete Schedule.		

2.2.15 Import image result

Table 2.2.15.1 Import image result use case

Primary actor:	Receptionist	Secondary actor:	None
Description:	Allow the receptionist to import result image to medical notebook of patient		
Trigger:	User click “+” on the side of schedule information row		
Pre-condition:	The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. Receptionist click on “Sổ tay khám bệnh”, then the system display all of medical notebook. 2. Receptionist clicks on “+” to import the image. 3. Click on “Lưu” to save change.		
Alternative flows:	2.1. Users click “X” on the popup notification. 2.2 The system cancelled the import image.		

2.2.16 View Feedback

Table 2.2.16.1 View Feedback use case

Primary actor:	All user	Secondary actor:	None
Description:	Allow all user to view the feedback of patients		
Trigger:	1. The User clicks button “feedback”		
Pre-condition:	1. The user is logged into the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. The user click on button feedback 2. The system displays the feedbacks of patients as a list : comment , rate ,...		
Alternative flows:	None		

2.2.17 View List Employee Account

Table 2.2.17.1 View List Employee Account use case

Primary actor:	Administrators	Secondary actor:	None
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Description:	The purpose of this use case is to allow administrators to view a list of employee accounts in the system
Trigger:	1. The User click button “Danh sách nhân viên”
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.
Post-condition:	None
Normal flow:	1. The User click button “Danh sách nhân viên” 2. The system display list of employee account, but divided in 2 sections: Doctor and Receptionist, the user need to choose 1 of this 3. The system display showing account information such as name, ID corresponding to Doctor or Receptionist
Alternative flows:	None

2.2.18 Add Employee Account

Table 2.2.18.1 Add Employee Account use case

Primary actor:	Administrators	Secondary actor:	None
Description:	The purpose of this use case is to allow administrators to create accounts for employees to into system		
Trigger:	1. User click “Tạo tài khoản” to create new account for employee		
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	1. User adds a new employee account to system and display it in table of list employee account		
Normal flow:	1. User clicks the “Tạo tài khoản” button then the system will redirect the user to create a page to enter a new employee. 2. Users enter employee information and then click the “Lưu” button to create a new employee. 3. The system validates the employee input information. 4. System added a new employee. 5. System notify users of successful action.		
Alternative flows:	3.1 If the user does not enter required information fields such as name, code... The system will notify you of missing information and ask the user to enter all required information to create a new		

	employee account
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2.2.19 Change status of Receptionist(isActive)

Table 2.2.19.1 Change status of Receptionist(isActive) use case

Primary actor:	Administrators	Secondary actor:	None
Description:	Allow the admin to change doctor status, if isActive = true, that mean this account has permission and active		
Trigger:	User click “Thay đổi” on the side of employee information row		
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. The system show list of Receptionist and action change beside the Receptionist information 2. The user clicks “Thay đổi” and “Xác nhận” on the popup show to confirm the change schedule. 3. The system updates the status of Receptionist . 4. The system notified users of successful action.		
Alternative flows:	2.1. User click “Hủy” on the popup notification. 2.2 The system cancelled the change of employee information.		

2.2.20 Set Status for patient

Table 2.2.20.1 Set Status for patient use case

Primary actor:	Receptionist	Secondary actor:	None
Description:	The purpose of this use case is to allow the receptionist to enter patient information for examination		
Trigger:	1. User click on “Tạo xét nghiệm” to create new request for examination and enter patient information		
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. User “Tạo xét nghiệm” and the system show up form to enter		

	patient information for examination 2. User enter the information of patent into form
Alternative flows:	None

2.2.21 Create Patient Account

Table 2.2.21.1 Create Patient Account use case

Primary actor:	Receptionist	Secondary actor:	None
Description:	The purpose of this use case is to allow the receptionist to provide an account to the patient		
Trigger:	1. User clicks the “Tạo tài khoản bệnh nhân” button then the system will redirect the user to create a page to enter a new patient. 2. Users enter patient information and then click the “Tạo tài khoản” button to create a new patient. 3. The system validates the patient input information. 4. System added a new patient. 5. System notify users of successful action.		
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	1. User adds a new patient account to system and display it in table of list patient account		
Normal flow:	1. Click on the button “Tạo tài khoản bệnh nhân” and system will redirect user to create page. 2. User enter patient information and click “Tạo tài khoản”. 3. The system validates the information that users enter and creates a request. 4. A popup show that creates requests successfully.		
Alternative flows:	3.1. If the user does not enter required information fields such as name, code... The system will notify you of missing information and ask the user to enter all required information to create a new request for examination.		

2.2.22 Answer Feedback

Table 2.2.22.1 Answer Feedback use case

Primary actor:	Receptionist	Secondary actor:	Patient
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Description:	The purpose of this use case is to allow users to respond to feedback
Trigger:	1. User view the feedback and click “Trả lời” to answer the feedback
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.
Post-condition:	1. Users answer the feedback
Normal flow:	1. Patients create the feedback. 2. Receptionist views the list of feedback and clicks on “Trả lời” to answer the patient's feedback. 3. Patient can view the answer of receptionist and also can click on “Trả lời” to answer receptionist answer
Alternative flows:	3.1. If the user does not enter required information fields such as name, code... The system will notify you of missing information and ask the user to enter all required information to create a new request for examination.

2.2.23 Approve Appointment Information

Table 2.2.23.1 Approve Appointment Information use case

Primary actor:	Receptionist	Secondary actor:	None
Description:	The purpose of this use case is to allow the receptionist to approve a patient's appointment request		
Trigger:	1. Users view the list of appointments and click “Xác nhận” if they approve the appointment. 2. In case of declining appointments, the user clicks on “Từ chối”.		
Pre-condition:	1. The user is currently logged in to the system and has the corresponding permissions.		
Post-condition:	1. User approve of decline the appointment		
Normal flow:	1. User view list of appointment and take action that to approve or decline that appointment by clicks on “Xác nhận” or “Từ chối” button.		
Alternative flows:	None		

2.2.24 Register

Table 2.2.24.1 Register use case

Primary actor:	Patient	Secondary actor:	None
Description:	This use case describes the process by which a new patient registers for an account in the healthcare system.		
Trigger:	1. The patient decides to register for an account to access personalised healthcare services.		
Pre-condition:	1. The patient is not currently registered in the system.		
Post-condition:	1. The patient's account is successfully created in the system.		
Normal flow:	1. The patient logs in. 2. Redirected to the homepage. 3. The patient fills out the required information, including title, the problem they are facing. 4. The patient submits the question form. 5. The question will be sent to the doctor. 6. The doctor receive the question, give an answer and deliver to the patient		
Alternative flows:	None		

2.2.25 Ask doctor a question

Table 2.2.25.1 Ask doctor a question use case

Primary actor:	Patient	Secondary actor:	Doctor
Description:	This use case describes the process by which a patient gives a question to a doctor.		
Trigger:	1. The patient select “Đặt câu hỏi” to the doctor.		
Pre-condition:	1. The patient is not currently registered in the system.		
Post-condition:	1. The patient's account is successfully created in the system.		
Normal flow:	1. The patient accesses the registration page of the MedPal system. 2. The system displays the registration form. 3. The patient fills out the required information, including personal details (such as name, date of birth, contact information) and login credentials (such as username and password). 4. The patient submits the registration form.		

	- The system validates the entered information. - If all information is valid, the system creates a new account for the patient. - The system confirms the successful registration and provides the patient with a confirmation message or email.
Alternative flows:	5a. If the entered information is invalid or incomplete, the system displays error messages prompting the patient to correct the errors.

2.2.26 Update question

Table 2.2.26.1 Update question use case

Primary actor:	Patient	Secondary actor:	Doctor
Description:	This use case describes the process by how a patient can update the question they sent to the doctor.		
Trigger:	1. The patient selects to “Hỏi đáp chuyên gia”		
Pre-condition:	1. The patient must be logged in.		
Post-condition:	1. The question has been updated.		
Normal flow:	- The patient logs in. - Redirected to the homepage. - The patient takes a look for the question they sent before. - The patient can edit the question’s contents and select edit. - The question will be re-sent to the doctor. - The doctor receives the question, gives an answer and delivers it to the patient.		
Alternative flows:	None		

2.2.27 Update answer

Table 2.2.27.1 Update answer use case

Primary actor:	Doctor	Secondary actor:	Patient
Description:	This use case describes the process by how a doctor can update the question they sent to the doctor.		
Trigger:	1. The doctor selects to edit icon		

Pre-condition:	1. The doctor must be logged in.
Post-condition:	1. The answer has been updated.
Normal flow:	1. The doctor logs in. 2. Redirected to the homepage. 3. The doctor takes a look for the answer they sent before. 4. The doctor can edit the answer and select edit. 5. The answer will be re-sent to the patient.
Alternative flows:	None

2.2.28 View List of Doctors

Table 2.2.28.1 View List of Doctors use case

Primary actor:	All users	Secondary actor:	None
Description:	This use case allows a patient to view a list of available doctors within the healthcare system, including details such as specialty, location, and availability.		
Trigger:	1. The patient wants to find and select a doctor for a medical consultation or treatment.		
Pre-condition:	1. The patient has access to the healthcare system.		
Post-condition:	1. The patient can view a comprehensive list of doctors and access detailed information about each doctor.		
Normal flow:	1. The patient navigates to the "Danh sách bác sĩ" section of the healthcare system. 2. The system displays a list of available doctors. 3. The patient uses search and filter options to refine the list based on criteria such as specialty, location, or doctor's name. 4. The patient selects a doctor from the list to view detailed information about the doctor. 5. The system displays the selected doctor's details, including specialty, experience, availability, and contact information.		
Alternative flows:	3a. The patient refines the search criteria to better match their specific needs (e.g., insurance coverage, language spoken).		

2.2.29 View Doctor's Information

Table 2.2.29.1 View Doctor's Information use case

Primary actor:	All Users	Secondary actor:	None
Description:	This use case allows a patient to view detailed information about a specific doctor, including their specialty, qualifications, experience, availability, and contact details.		
Trigger:	The patient selects a doctor from the list of available doctors to view more detailed information.		
Pre-condition:	• The patient has access to the healthcare system. • The patient has selected a doctor from the list of available doctors.		
Post-condition:	• The patient can view comprehensive information about the selected doctor.		
Normal flow:	1. The patient navigates to the "Danh sách bác sĩ" section and selects a doctor from the list. 2. The system retrieves and displays detailed information about the selected doctor 3. The system retrieves and displays detailed information about the selected doctor.		
Alternative flows:	None		

2.2.30 View List Service

Table 2.2.30.1 View List Service use case

Primary actor:	All Users	Secondary actor:	None
Description:	This function allows users to see the entire list of services at the clinic and has filters so users can search according to their needs.		
Trigger:	1. The patient click on “ dịch vụ y tế ”		
Pre-condition:	1. The patient has access to the healthcare system.		
Post-condition:	1. The patient receives all of the service in the clinic and relevant search results based on their query.		
Normal flow:	1. The patient click on “ dịch vụ y tế ” 2. The system display all of services services in clinic		
Alternative flows:	1. The patient choose item in the filter like : Location,		

	examination list , ... and click on “Tìm kiếm ” 2. The system show all the service depend on the filter the patient choose
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2.2.31 View service detail

Table 2.2.31.1 View service detail use case

Primary actor:	All users	Secondary actor:	None
Description:	This use case allows a patient to view a comprehensive list of services offered by the healthcare facility, including descriptions, pricing, and availability.		
Trigger:	1. Users select a list of services.		
Pre-condition:	1. The patient has access to the healthcare system.		
Post-condition:	1. The patient can view detailed information about the services available.		
Normal flow:	1. The patient navigates to the "Services" section of the healthcare system. 2. The system displays a list of available services. 3. The patient can browse through the list or use search and filter options to find specific services. 4. The patient selects a service to view more detailed information. 5. The system displays detailed information about the selected service		
Alternative flows:	None		

2.2.32 Search Blog

Table 2.2.32.1 Search Blog use case

Primary actor:	All users	Secondary actor:	None
Description:	This use case allows a patient to search for blog posts within the healthcare system's blog, using keywords, categories, or other filters to find relevant health information articles.		
Trigger:	1. The patient wants to find specific health information or articles within the healthcare system's blog.		
Pre-condition:	1. The patient has access to the healthcare system.		

Post-condition:	1. The patient can view a list of blog posts that match the search criteria.
Normal flow:	1. The patient enters keywords or selects filters (e.g., categories, tags) in the search bar. 2. The patient initiates the search. 3. The system processes the search query and retrieves blog posts that match the search criteria. 4. The system displays a list of blog posts that match the search criteria. 5. The patient reviews the search results and selects a blog post to read more details.
Alternative flows:	None

2.2.33 View List of Blog Groups

Table 2.2.33.1 View List of Blog Groups use case

Primary actor:	All users	Secondary actor:	None
Description:	This use case allows a patient to view a list of blog groups within the healthcare system, categorized by different health topics or themes.		
Trigger:	1. The patient wants to explore blog posts organized by specific health topics or themes.		
Pre-condition:	1. The patient has access to the healthcare system.		
Post-condition:	1. The patient can view a list of blog groups and access blog posts within each group.		
Normal flow:	1. The patient navigates to the "Blog" section of the healthcare system. 2. The patient selects the option to view blog groups. 3. The system retrieves and displays a list of available blog groups. 4. The patient browses through the list of blog groups. 5. The patient selects a blog group to view the blog posts within that group. 6. The system displays the blog posts that belong to the selected blog group.		
Alternative flows:	3a. If the patient uses search or filter options to find specific blog groups, the system refines the list based on the criteria.		

2.2.34 Create Feedback

Table 2.2.34.1 Create Feedback use case

Primary actor:	Patient	Secondary actor:	None
Description:	This use case allows a patient to provide feedback on their experience with the healthcare system or services received.		
Trigger:	1. After the examination, the receptionist saves the examination results, sets the status to offline and the patient logs into the system for the first time after the examination.		
Pre-condition:	1. Receptionist set patient offline 2. The patient is logged into the healthcare system.		
Post-condition:	1. The feedback is submitted and recorded in the system. 2. A notification confirming the feedback submission may be sent to the patient.		
Normal flow:	1. Patient logs in for the first time after last visit 2. Patients leave comments about clinic services 3. Patients evaluate objectively based on star rating.		
Alternative flows:	4a. If the patient decides not to leave the feedback, patient can not access to make any action on website		

2.2.35 Book Appointment

Table 2.2.35.1 Book Appointment use case

Primary actor:	Patient	Secondary actor:	Receptionist
Description:	This use case allows a patient to create a new medical appointment with a doctor through the healthcare system.		

Trigger:	1. The patient needs to schedule a medical consultation or treatment with a doctor.
Pre-condition:	1. The patient is logged into the healthcare system. 2. Patients select by step: department, services, day by doctor schedule, time by doctor time.
Post-condition:	1. The appointment is successfully created and recorded in the system. 2. Notifications are sent to the patient and the receptionist confirming the appointment.
Normal flow:	1. The patient navigates to the "Đặt lịch khám" section of the healthcare system. 2. The patient selects the option to create a new appointment. 3. The system displays a form for creating an appointment, including available doctors, dates, and times. 4. The patient selects department, services, a doctor and chooses an available date and time. 5. The patient submits the appointment request. 6. The system validates the information and checks for any scheduling conflicts. 7. If no conflicts are found, the system creates the appointment and updates the doctor's schedule.
Alternative flows:	4a. If the patient cannot find a suitable time, they can modify their search criteria or select a different doctor. 8a. If a scheduling conflict is found, the system prompts the patient to select a different time or date. 1. Receptionist looking for patient needing appointment 2. The receptionist clicks on the "Tái " button and selects the department, service, doctor and date and time according to the patient's request. 3. Receptionist presses button "Đặt lịch"

2.2.36 View Appointment

Table 2.2.36.1 View Appointment use case

Primary actor:	Patient	Secondary actor:	None
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Description:	This use case allows a patient to view details of their scheduled appointments within the healthcare system.
Trigger:	1. The patient accesses the healthcare system to check details of their upcoming appointments or past appointment history.
Pre-condition:	1. The patient is logged into the system.
Post-condition:	1. The patient successfully views the details of their appointments.
Normal flow:	1. The patient navigates to the "Xem lịch khám" section of the healthcare system. 2. The system displays a list of the patient's upcoming appointments and/or past appointment history. 3. The patient selects a specific appointment from the list to cancel the appointment if they want. 4. The system retrieves and displays detailed information about the selected appointment
Alternative flows:	3a. If the patient has no upcoming appointments or past appointment history, the system show the table with content “Không có cuộc hẹn nào” 3b. If the patient wants to view a specific type of appointment (e.g., upcoming appointments only, past appointments), the system provides options to filter the appointment list accordingly.

2.2.37 Cancel Appointment

Table 2.2.37.1 Cancel Appointment use case

Primary actor:	Patient	Secondary actor:	None
Description:	This use case allows a patient to cancel a scheduled appointment within the healthcare system.		
Trigger:	1. The patient can choose if they have a reason for not visiting the appointment.		
Pre-condition:	1. The patient is logged into the healthcare system. 2. The patient has an existing appointment that they want to cancel.		
Post-condition:	1. The appointment is successfully cancelled or deleted from the system. 2. Notifications are sent to the patient and the doctor to inform		

	them of the cancelled appointment.
Normal flow:	1. The patient navigates to the "Xem lịch khám" section of the healthcare system. 2. The system displays a list of the patient's upcoming appointments. 3. The patient selects the appointment they want to cancel from the list. 4. The system displays detailed information about the selected appointment, including the option to cancel. 5. The patient confirms the cancellation of the appointment. The system will immediately delete appointment if appointment haven't been approved yet. 6. The system sends a notification to the patient and the doctor to inform them of the cancelled appointment.
Alternative flows:	3a. If the patient has no upcoming appointments, the system displays a message "Không có cuộc hẹn nào". 4a. If the appointment has already passed, the system may still allow the patient to cancel it but notifies the patient that the appointment has already occurred. 5a. If appointment have been approved the "Trạng thái" column will change to "Đã hủy"

2.2.38 Receive of Follow-Up Appointment

Table 2.2.38.1 Receive of Follow-Up Appointment use case

Primary actor:	Patient	Secondary actor:	None
Description:	The purpose of this use case is to describe how the system reminds patients for follow-up appointments .		
Trigger:	1. The appointment date is approaching (2-3 days before the scheduled follow-up appointment).		
Pre-condition:	1. The patient has a scheduled follow-up appointment in the system with contact details (SMS or email).		
Post-condition:	1. The patient receives a reminder notification via SMS or email.		
Normal flow:	1. The system checks the schedule daily to identify follow-up appointments occurring in the next 2-3 days 2. The system retrieves the patient's contact information (SMS or email) from the database. 3. The system generates a reminder message.		

	- The system sends the reminder message via SMS or email to the patient. - The patient receives the reminder and prepares for the follow-up appointment.
Alternative flows:	None

2.2.39 Write Blog Posts about Health Information

Table 2.2.39.1 Write Blog Posts about Health Information use case

Primary actor:	Article manager	Secondary actor:	None
Description:	The purpose of this use case is to allow the Article managers to write blog posts about Health Information		
Trigger:	1. The Article manager click “Thêm mới” button		
Pre-condition:	- The Article manager is currently logged in to the system - The Article manager’s account has been authorised - The Article manager's device is connected to the internet when logging in		
Post-condition:	1. The blog post will be uploaded to the system and be shown.		
Normal flow:	- The Article manager fills all the fields and clicks on the “Thêm” button to upload the word file(the word file will become blog content). - The Article click on “Thêm mới” button to upload post.		
Alternative flows:	None		

2.2.40 Delete Blog Posts about Health Information

Table 2.2.40.1 Delete Blog Posts about Health Information use case

Primary actor:	Article manager	Secondary actor:	None
Description:	This use case describes how Article managers remove a blog post about Health Information From the system.		
Trigger:	1. The Article manager click on “Xóa” button		
Pre-condition:	- The Article manager is currently logged in to the system - The Article manager’s account has been authorised		

	3. The Article manager's device is connected to the internet when logging in
Post-condition:	1. The post will be deleted from the system and redirected to the homepage.
Normal flow:	1. The Article manager selects the article to remove. 2. The Article manager re-checks all the fields and selects the “Xóa” button.
Alternative flows:	None

2.2.41 View Blog Posts about Health Information

Table 2.2.41.1 View Blog Posts about Health Information use case

Primary actor:	All users	Secondary actor:	None
Description:	This use case describes the process and associated interactions when a user is willing to see a blog post about Health Information for a clinic.		
Trigger:	1. The User clicks on the wanted post.		
Pre-condition:	1. The user's device is connected to the internet		
Post-condition:	1. The blog post will be shown.		
Normal flow:	1. The system shows the blog post when clicking the title. 2. The user can use the search sidebar. The user can search by title of the blog posts.		
Alternative flows:	2.1 If users search with an empty string, it will show a list of latest blog post		

2.2.42 Update Blog Posts about Health Information

Table 2.2.42.1 Update Blog Posts about Health Information use case

Primary actor:	Article manager,Admin	Secondary actor:	None
Description:	The purpose of this use case is to allow Article managers to edit blog posts and upload it to the system.		
Trigger:	The Article manager clicks on the “Chỉnh sửa” button.		
Pre-condition:	1. The Article manager is currently logged in to the system		

	- The Article manager's account has been authorised - The Article manager's device is connected to the internet when logging in
Post-condition:	The blog post will be updated after finishing.
Normal flow:	- The Article manager selects the article to edit. - The Article manager checks all the fields, chooses which fields to edit, then clicks on the "Edit" button to edit the blog post.
Alternative flows:	2.1 If The Article manager left an empty field, it will show a reminder not to leave empty.

2.2.43 Create Medical Notebook

Table 2.2.43.1 Create Medical Notebook use case

Primary actor:	Doctor	Secondary actor:	None
Description:	The purpose of this use case is to allow Doctor to create medical notebook		
Trigger:	The Doctor clicks on the "Tạo sổ tay khám bệnh" button.		
Pre-condition:	- The Doctor is currently logged in to the system - The Doctor's account has been authorised - The Doctor's device is connected to the internet when logging in		
Post-condition:	Add new medical notebook to database		
Normal flow:	- Doctor click on "Xem bệnh nhân đang hoạt động" - Doctor clicks on button "Thêm mới" - Doctor fills all the field and click "Tạo"		
Alternative flows:	None		

2.2.44 Reminder follow-up appointment

Table 2.2.44.1 Reminder follow-up appointment use case

Primary actor:	Admin	Secondary actor:	None
Description:	The purpose of this use case is to allow Admin to remind follow-up appointment to patient		
Trigger:	The Admin clicks on the "Xem lịch sử khám của bệnh nhân" in		

	sidebar and click on “chỉnh sửa thời gian nhắc tái khám”
Pre-condition:	1. The Admin is currently logged in to the system 2. The Admin account has been authorised 3. The Admin’s device is connected to the internet when logging in
Post-condition:	Admin set time for reminder appointment
Normal flow:	1. Admin click on “Xem lịch khám” 2. Admin click on button “NHẮC LỊCH” 3. Admin set time for the reminder follow-up appointment and save 4. The System send the notification for patient by email or phone number
Alternative flows:	None

2.2.45 Online advising

Table 2.2.45.1 Online advising use case

Primary actor:	Patient	Secondary actor:	None
Description:	The purpose of this use case is to allow Patient to chat with receptionist and ask what they want		
Trigger:	The Patient click on “Nhận tư vấn ngay” icon with title on the left of bottom screen		
Pre-condition:	1. Patient currently logged in to the system 2. The Patient account has been authorised 3. The Patient device is connected to the internet when logging in		
Post-condition:			
Normal flow:	1. Patient click on icon with title “Nhận tư vấn ngay” 2. The system display the message box on the bottom right of the screen 3. Patient enter text and start chat with receptionist		
Alternative flows:	1. If there’s something that receptionist cant handle, the receptionist will be connect to the doctor and let them chat with patient		

2.2.46 Post treatment follow-up

Table 2.2.46.1 Post treatment follow-up use case

Primary actor:	Doctor	Secondary actor:	None
Description:	The purpose of this use case is to allow Doctor to chat and take care of patient after they examined		
Trigger:	The Doctor clicks on the “Tương tác với các bệnh nhân đã khám” on the sidebar.		
Pre-condition:	1. TheDoctor is currently logged in to the system 2. The Doctor’s account has been authorised 3. The Doctor's device is connected to the internet when logging in		
Post-condition:			
Normal flow:	1. Click on “Tương tác với các bệnh nhân đã khám” on the sidebar. 2. The system display all patient that they have examined 3. Click on “Chat” will appear as messenger on the web and they can chat with patient		
Alternative flows:	None		

2.2.47 View Doctor Schedule

Table 2.2.47.1 View Doctor Schedule use case

Primary actor:	Doctor	Secondary actor:	None
Description:	Allows view the doctor’s schedule in a week		
Trigger:	2. The User clicks button “Xem lịch làm việc hàng tuần”		
Pre-condition:	2. The user is logged into the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	5. The User clicks button “Xem lịch làm việc hàng tuần” 6. The user choose start date and end date 7. Click on “Gửi” to view schedule that match with start date and end date 8. The system displays the Schedule of doctors, showing information of schedule such as time, slot, ID..		

Alternative flows:	None
--------------------	------

2.2.48 View introduction article

Table 2.2.48.1 View introduction article use case

Primary actor:	All User	Secondary actor:	None
Description:	Allows view the doctor's schedule in a week		
Trigger:	3. The User clicks button "Xem lịch làm việc"		
Pre-condition:	3. The user is logged into the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	9. The User clicks button "Xem lịch làm việc" 10. The user choose start date and end date 11. Click on "Gửi" to view schedule that match with start date and end date 12. The system displays the Schedule of doctors, showing information of schedule such as time, slot, ID..		
Alternative flows:	None		

2.2.49 Search Service

Table 2.2.49.1 Search Service use case

Primary actor:	All User	Secondary actor:	None
Description:	Allows view the doctor's schedule in a week		
Trigger:	4. The User clicks button "Xem lịch làm việc"		
Pre-condition:	4. The user is logged into the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	13. The User clicks button "Xem lịch làm việc" 14. The user choose start date and end date 15. Click on "Gửi" to view schedule that match with start date and end date		

	16. The system displays the Schedule of doctors, showing information of schedule such as time, slot, ID..
Alternative flows:	None

2.2.50 Schedules follow-up appointment

Table 2.2.50.1 Schedules follow-up appointment use case

Primary actor:	Receptionist	Secondary actor:	None
Description:	Allows receptionist book a follow-up appointment for patient		
Trigger:	1. The User clicks on the “Tài khoản bệnh nhân” section and clicks on the “Tái khám” button to book a follow-up appointment.		
Pre-condition:	1. The user is logged into the system and has the corresponding permissions.		
Post-condition:	None		
Normal flow:	1. The User clicks “Tài khoản bệnh nhân” section 2. The user clicks on “Tái khám” button 3. Choose appointment by patient will 4. Save appointment		
Alternative flows:	None		

3. Functional Requirements

3.1 Schedule Management

3.1.1 Create Appointment

Function Trigger:

- The user click on “Đặt lịch” in homepage.

Function Description:

- This function allow user to booking an appointment online.

Screen Layout:

Đặt lịch khám

Chuyên khoa *
Cardiology

Dịch vụ *
Heart Checkup

Bác sĩ *
Doctor User

Ngày *
07/16/2024

Thời gian *
10:00-11:00

ĐẶT LỊCH



Figure 3.1.1.1 Create Appointment

Function Details:

Normal cases

- User choose “Đặt lịch khám” in homepage, fill all required information, and click on “Đặt lịch”.

Abnormal cases

- If the user does not fill in all required fields, they will not be able to press "Đặt lịch" and schedule.

3.1.2 View Appointment (Patient)

Function Trigger:

- Patient click on “Xem lịch khám”.

Function Description:

- The purpose of this use case describes how patients can view booked appointments.

Screen Layout:

Danh sách cuộc hẹn						
						ĐẶT LỊCH KHÁM Tất cả
Thời gian	Ngày	Bác sĩ	Dịch vụ	Trạng thái	Ghi chú	Hành động
10:00-11:00	15-08-2024	Nguyễn Thị Hoa	Tiểu hóa	Đang chờ phê duyệt		
16:00-17:00	15-08-2024	Nguyễn Đức Anh	Chụp X Quang	Đang chờ phê duyệt		
09:00-10:00	15-08-2024	Ngô Anh Duy	Sieu âm	Đang chờ phê duyệt		
14:00-15:00	15-08-2024	Nguyễn Thanh Dân	Hô hấp	Đang chờ phê duyệt		
10:00-11:00	15-08-2024	Nguyễn Đức Thịnh	Mổ nội soi cắt đại tràng	Đang chờ phê duyệt		

Figure 3.1.2.1 View Appointment (Patient)

Function Details:

Normal cases

- When the patient clicks on "Xem lịch khám", the table will be displayed as below. You can use the interactive buttons "Chỉnh sửa" and "Hủy lịch hẹn" to have the system display the corresponding popup.

Abnormal cases

- None.

3.1.3 Cancel Appointment

Function Trigger:

- Patient clicks on the bin icon.

Function Description:

- The purpose of this use case describes how patients can cancel their booking appointment.

Screen Layout:

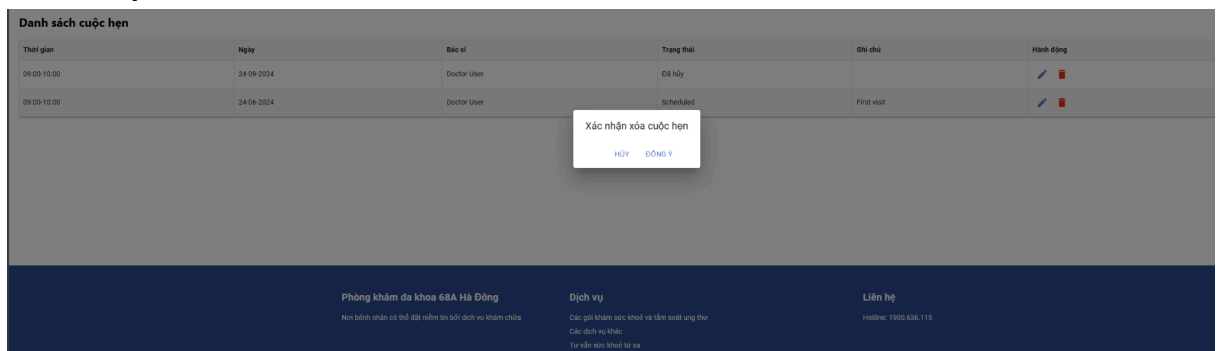


Figure 3.1.3.1 Cancel Appointment

Function Details:

Normal cases

- When the patient clicks on the delete icon, the Popup will be displayed as above to confirm cancel.
- if the status is “đang chờ phê duyệt” the system will delete the appointment
- if the status is “đã phê duyệt” the system will change appointment’s status to “Đã Hủy”

Abnormal cases

- if case except(“đã phê duyệt”, “đã hủy”) not have the delete icon

3.1.4 Approve the appointment request

Function Trigger:

- The receptionist selects “Phê duyệt” the appointment request.

Function Description:

- The purpose of this use case is to view the doctor’s schedule in a week.

Screen Layout:

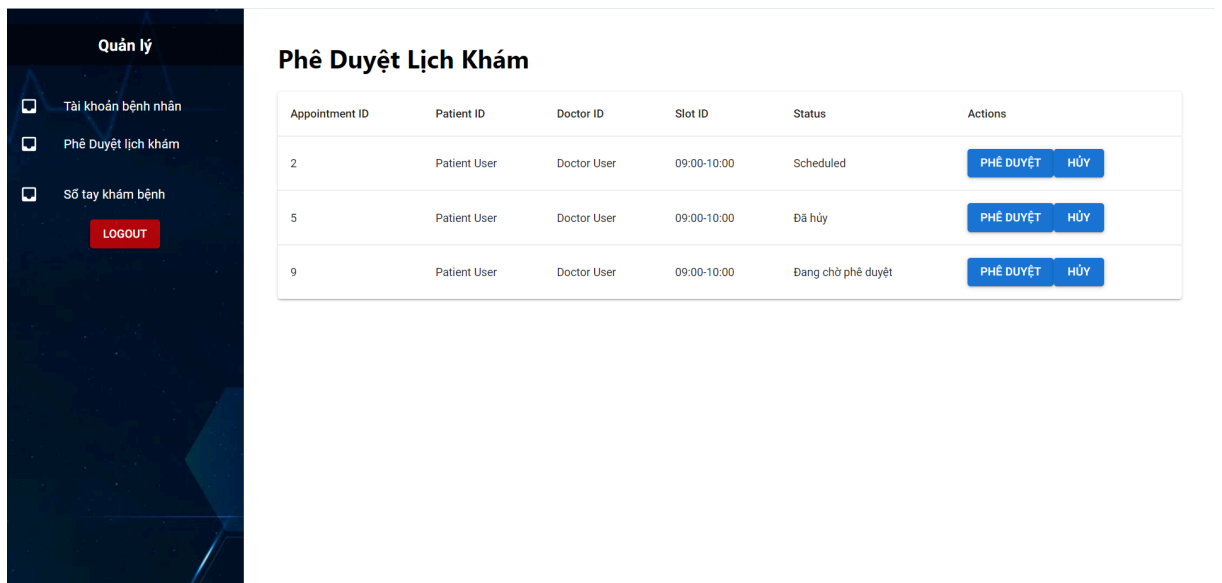


Figure 3.1.4.1 Approve the appointment request

Function Details:

Normal cases:

- When there is a new appointment request, the admin selects the “Phê Duyệt” button to approve the appointment request.

Abnormal cases:

- If the admin selects “Hủy” button to reject the appointment request.

3.1.5 Create Doctor Schedule

Function Trigger:

- The admin clicks on “Lịch làm việc” on the sidebar and selects doctor and fills choose date and schedule then saving by clicking on “Lưu lịch làm việc”.

Function Description:

- The purpose of this use case is to allow the admin to create a Schedule for doctors.

Screen Layout:

Quản lý lịch làm việc của bác sĩ

[THÊM MỚI LỊCH LÀM VIỆC](#)
[CHỈNH SỬA LỊCH LÀM VIỆC](#)
[XÓA LỊCH LÀM VIỆC](#)
[XEM LỊCH LÀM VIỆC](#)

Chọn bác sĩ

Doctor User

07/24/2024

07/30/2024

Lịch làm việc trong tuần

CN	T2	T3	T4	T5	T6	T7
KHÔNG	KHÔNG	KHÔNG	KHÔNG	KHÔNG	KHÔNG	KHÔNG
KHÔNG	KHÔNG	KHÔNG	KHÔNG	KHÔNG	KHÔNG	KHÔNG

LƯU LỊCH LÀM VIỆC

Figure 3.1.5.1 Create Doctor Schedule

Function Details:

Normal cases:

- The admin creates a new schedule and saves, the system will check input information, the schedule will be created if valid.

Abnormal cases:

- If the input information is invalid a new schedule won't be created.

3.1.6 Update Doctor Schedule

Function Trigger:

- The admin selects “Chỉnh sửa lịch làm việc” tab.

Function Description:

- The purpose of this use case is to allow the admin to update the Schedule for the doctor.

Screen Layout:

Quản lý lịch làm việc của bác sĩ

[THÊM MỚI LỊCH LÀM VIỆC](#)
 [CHỈNH SỬA LỊCH LÀM VIỆC](#)
 [XÓA LỊCH LÀM VIỆC](#)
 [XEM LỊCH LÀM VIỆC](#)

Chọn bác sĩ: ▼

Doctor User

← Tuần bắt đầu từ 24/06/2024 →

	Mon 24/06	Tue 25/06	Wed 26/06	Thu 27/06	Fri 28/06	Sat 29/06	Sun 30/06
Morning	CÓ LỊCH	-	-	-	-	-	-
Afternoon	NGHỈ	-	-	-	-	-	-

[LƯU LỊCH LÀM VIỆC](#)

Figure 3.1.6.1 Update Doctor Schedule

Function Details:

Normal cases:

- The admin selects the doctor, the system will show the list by week. Then change shift by clicking on the day and button. Finish by clicking on “Lưu lịch làm việc”.

Abnormal cases:

- If the input information is invalid the schedule won't be updated to the doctor.

3.1.7 Delete Doctor Schedule

Function Trigger:

- The admin selects the “Xóa lịch làm việc” tab.

Function Description:

- The purpose of this use case is to allow the admin to delete the Schedule for the doctor.

Screen Layout:

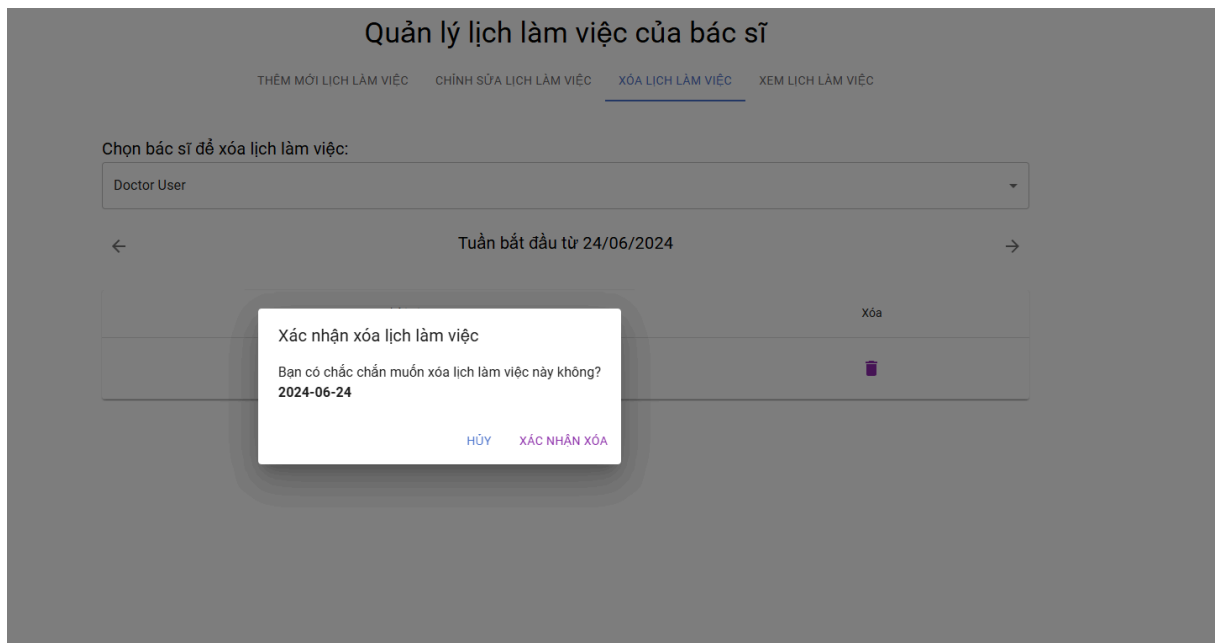


Figure 3.1.7.1 Delete Doctor Schedule

Function Details:

Normal cases:

- The admin selects the doctor, the working schedule will be shown as a list. If the admin wants to delete, then select the delete icon. A confirmation popup will be shown to ensure the delete.

Abnormal cases:

- if the admin don't want to remove the schedule, selecting "Hủy bỏ" to cancel deleting the schedule.

3.1.8 View Doctor Schedule(Admin)

Function Trigger:

- The user clicks on "Xem lịch làm việc" tab.

Function Description:

- The purpose of this use case is to allow viewing the doctor's schedule in a week.

Screen Layout:

Quản lý lịch làm việc của bác sĩ

[THÊM MỚI LỊCH LÀM VIỆC](#) [CHỈNH SỬA LỊCH LÀM VIỆC](#) [XÓA LỊCH LÀM VIỆC](#) [XEM LỊCH LÀM VIỆC](#)

Chọn bác sĩ

Doctor User

Ngày bắt đầu

01/16/2024

Ngày kết thúc

12/23/2024

GỬI

Ngày	Buổi Sáng	Buổi Chiều
24/06/2024	Có lịch	Nghỉ

Figure 3.1.8.1 View Doctor Schedule(Admin)

Function Details:

Normal cases:

- The user selects a specified schedule, showing information of schedule.

Abnormal cases:

- if the selected date is invalid the schedule won't be shown.

3.1.9 View Appointment

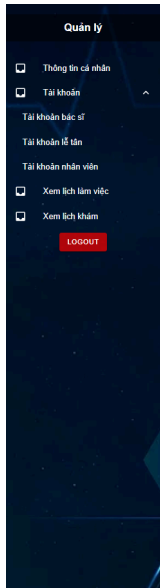
Function Trigger:

- Admin click on "Xem lịch khám".

Function Description:

- The purpose of this use case describes how Admin can view patient appointments.

Screen Layout:



ID	Tên bệnh nhân	Tên bác sĩ	Ngày	Thời gian	Trạng thái	Ghi chú
20	Lê Trung Dũng	Nguyễn Thị Hoa	08-08-2024	09:00-10:00	Tải khám	Không có
21	Ngô Kiến Huy	Nguyễn Thị Hoa	15-08-2024	10:00-11:00	Đang chờ phê duyệt	Không có
22	Ngô Kiến Huy	Nguyễn Đức Anh	15-08-2024	16:00-17:00	Đang chờ phê duyệt	Không có
23	Ngô Kiến Huy	Ngô Anh Duy	15-08-2024	09:00-10:00	Đang chờ phê duyệt	Không có
24	Ngô Kiến Huy	Nguyễn Thanh Dân	15-08-2024	14:00-15:00	Đang chờ phê duyệt	Không có
25	Ngô Kiến Huy	Nguyễn Đức Thịnh	15-08-2024	10:00-11:00	Đang chờ phê duyệt	Không có

Figure 3.1.9.1 View Appointment

Function Details:

Normal cases

- When the admin clicks on "Xem lịch khám", the table will be displayed as below.

Abnormal cases

- None.

3.1.10 View Doctor Schedule(Doctor)

Function Trigger:

- The user clicks on "Xem lịch làm việc hàng tuần" tab.

Function Description:

- The purpose of this use case is to allow viewing the doctor's schedule in a week.

Screen Layout:

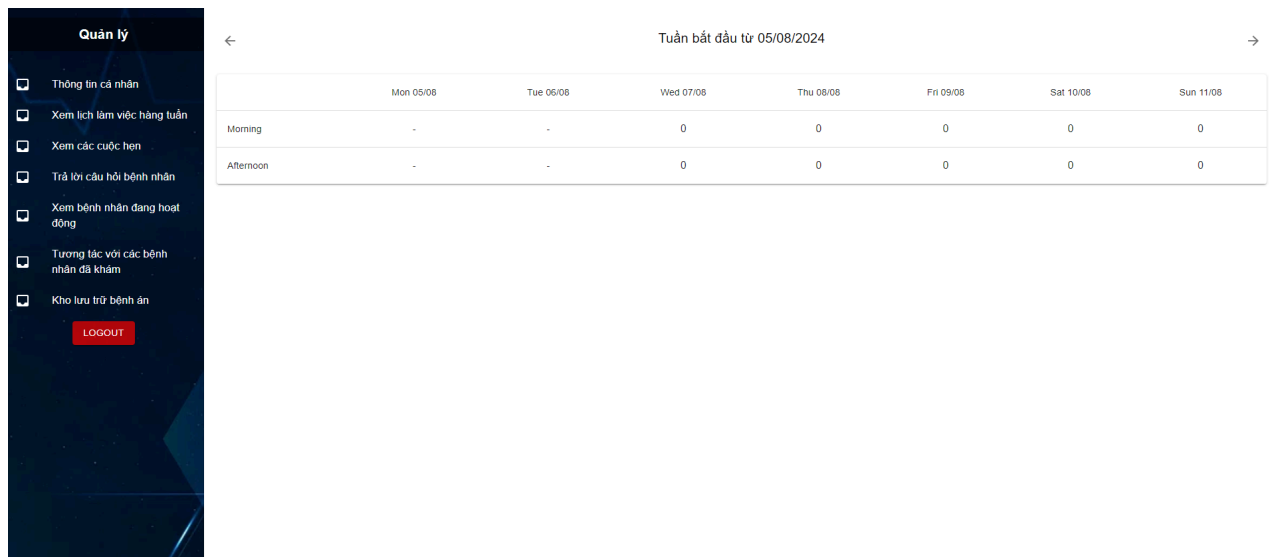


Figure 3.1.10.1 View Doctor Schedule(Doctor)

Function Details:

Normal cases:

- The doctor selects a week , the system will show the timetable of the week and the number of appointments that doctor has in the weekly's schedule.

Abnormal cases:

- none

3.2 Online Consulting

3.2.1 Reminder follow-up appointment(patient)

Function Trigger:

- The user login to email

Function Description:

- The purpose of this use case is to allow users to view notification about upcoming medical examination schedules

Screen Layout:

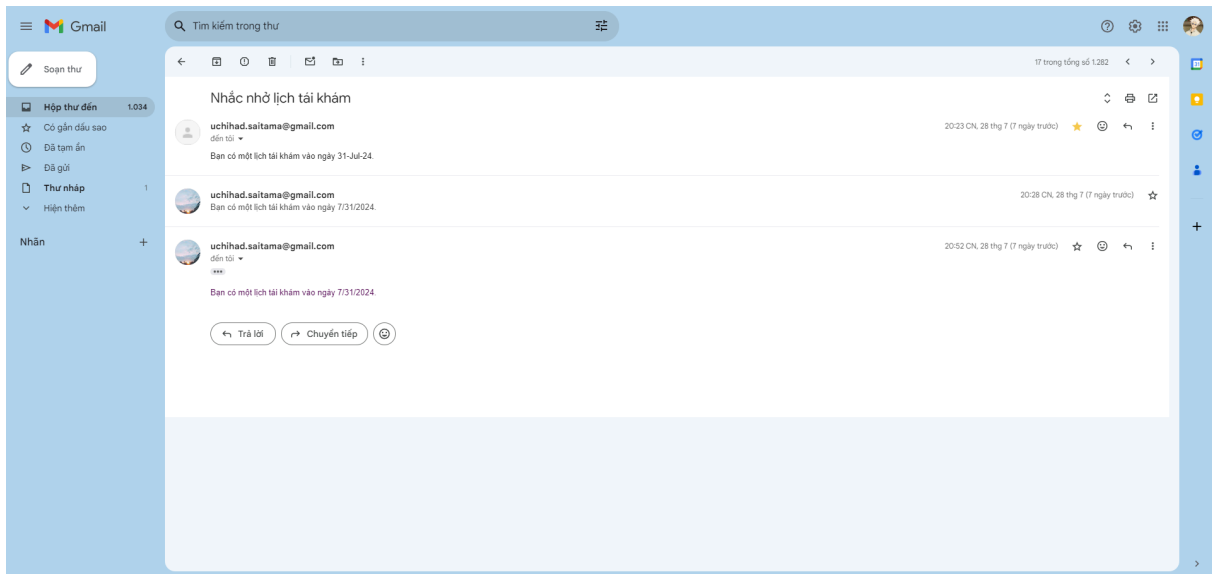


Figure 3.2.1.1 Reminder follow-up appointment(patient)

Function Details:

Normal cases

- Users can view view notification about upcoming medical examination schedules by login and see it through email

Abnormal cases

- none

3.2.2 Reminder follow-up appointment(admin)

Function Trigger:

- The user clicks on “Xem lịch sử khám của bệnh nhân ”in the sidebar

Function Description:

- The purpose of this use case is to allow users to view list of appointment and they can set time for reminder follow up appointment and the system will send notification to patient by email or phone number

Screen Layout:

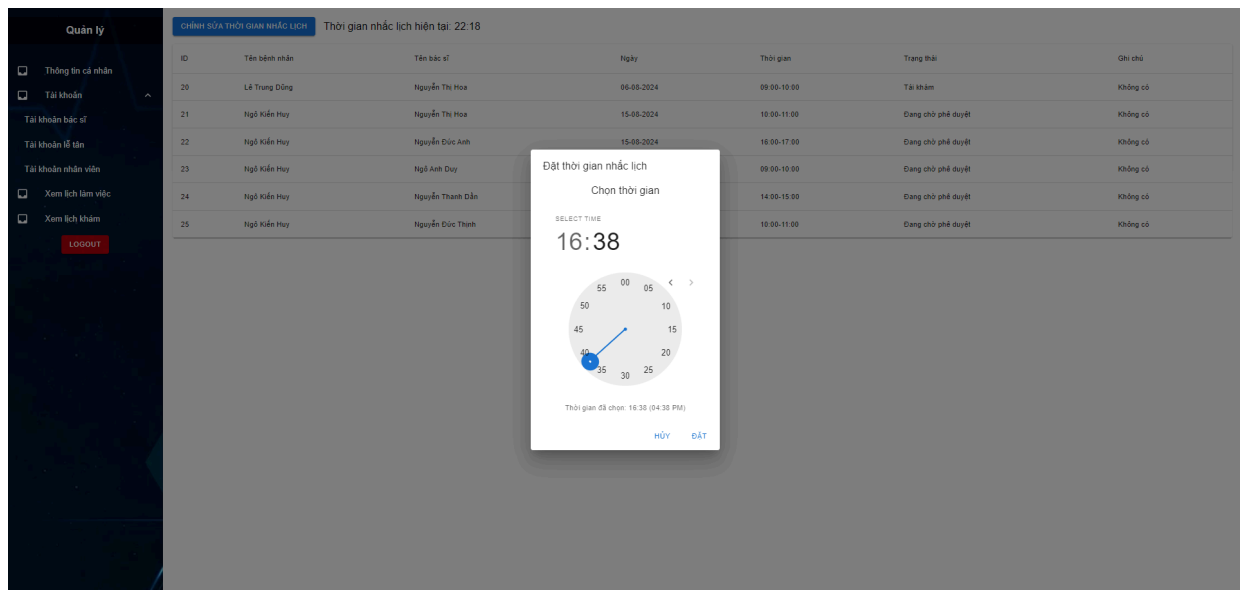


Figure 3.2.2.1 Reminder follow-up appointment(admin)

Function Details:

Normal cases

- Users can view history of appointment of patient and they can set time for reminder follow up appointment and the system will send notification to patient by email or phone number

Abnormal cases

- none

3.2.3 Post treatment follow-up(Doctor)

Function Trigger:

- The user clicks on “Tương tác với bệnh nhân đã khám” on sidebar

Function Description:

- The purpose of this use case is to allow doctor to chat with patient that they have examined

Screen Layout:

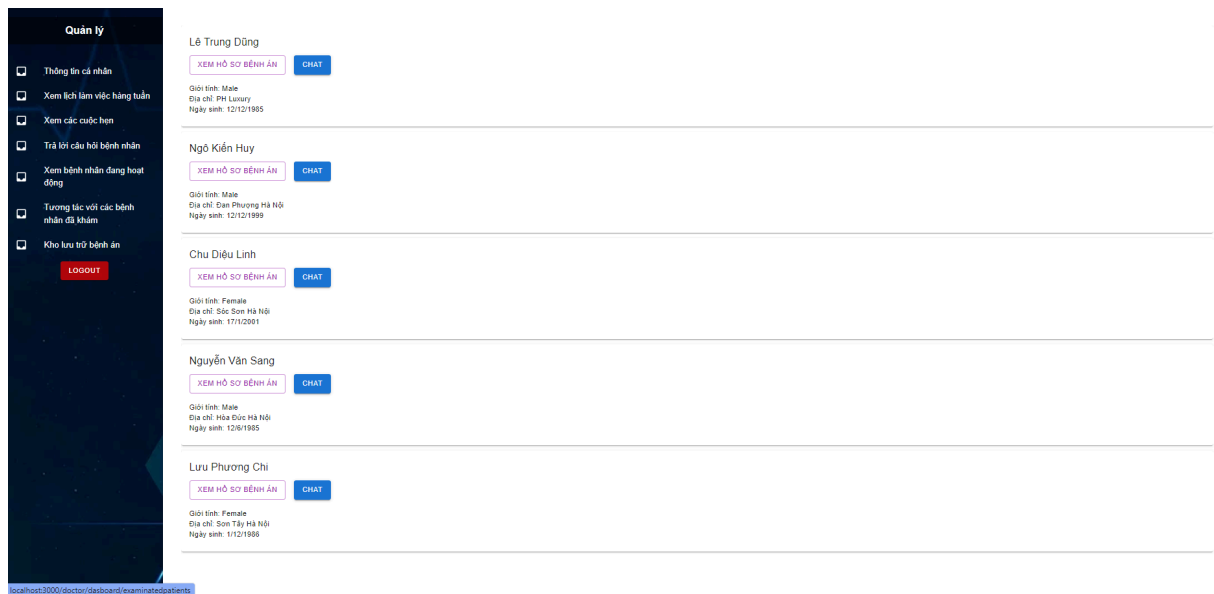


Figure 3.2.3.1 Post treatment follow-up(Doctor)

Function Details:

Normal cases

- Click on “Tương tác với bệnh nhân đã khám”
- The system display all patient that they have examined
- Click on “chat” will appear as messenger on the web and they can chat with patient

Abnormal cases

- None

3.2.4 Online advising(Patient)

Function Trigger:

- The user clicks on “Tư vấn online”

Function Description:

- The purpose of this use case is to allow users to chat with receptionist and ask what they want

Screen Layout:

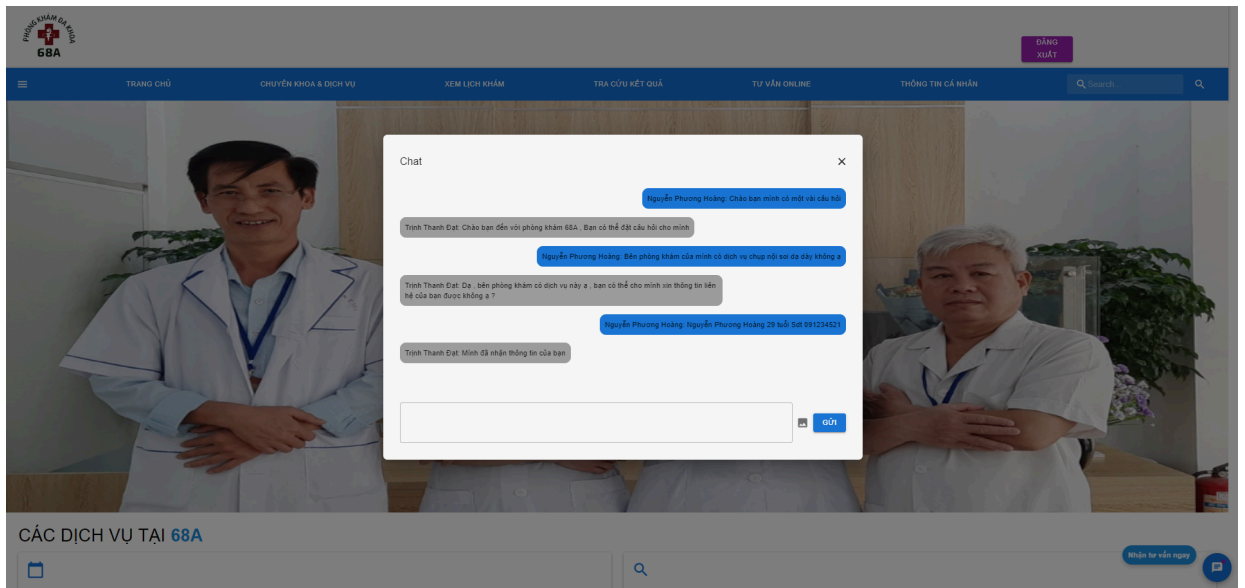


Figure 3.2.4.1 Online advising(Patient)

Function Details:

Normal cases

- Click on “Tự vấn online”
- System show message popup on the bottom right of the screen
- Patient enter text and start chat with receptionist

Abnormal cases

- None

3.2.5 Online advising(Receptionist)

Function Trigger:

- The user clicks on icon chat in the right-bottom corner

Function Description:

- The purpose of this use case is to allow receptionist to chat with patient

Screen Layout:

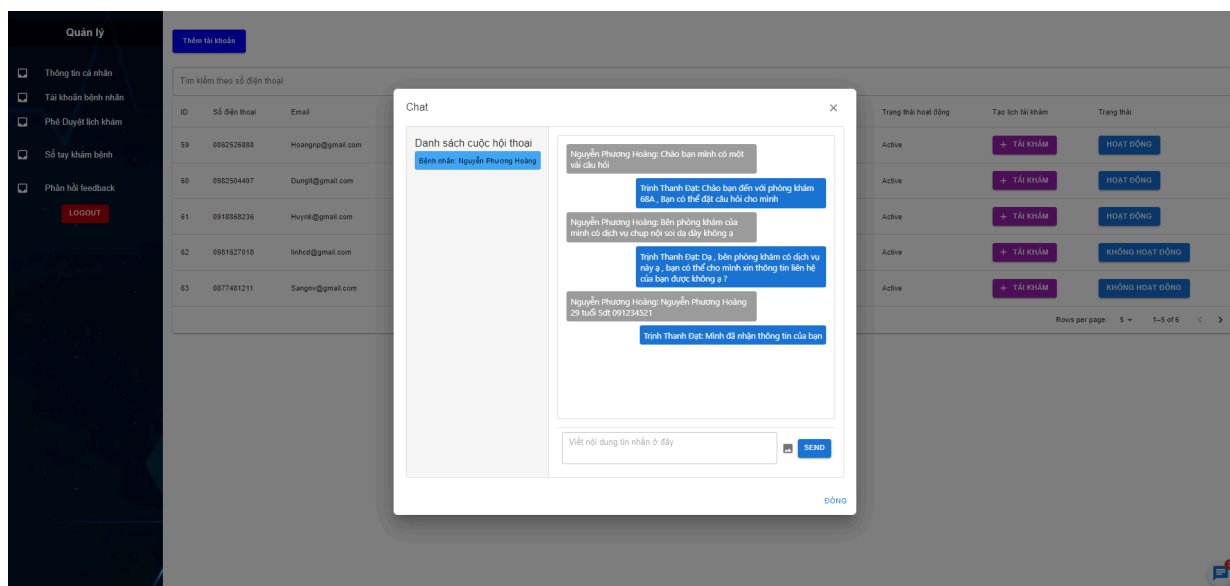


Figure 3.2.5.1 Online advising(Receptionist)

Function Details:

Normal cases

- Click on on icon chat in the right-bottom corner
- System show message popup on the bottom right of the screen
- receptionist enter text and start chat with patient

Abnormal cases

- None

3.3 Medical Notebooks management

3.3.1 Create Medical Notebook

Function Trigger:

- Doctor choose “Xem bệnh nhân đang hoạt động”

Function Description:

- This function is used to create new information (prescription, treatment) in the Medical Notebook of a patient.

Screen Layout:

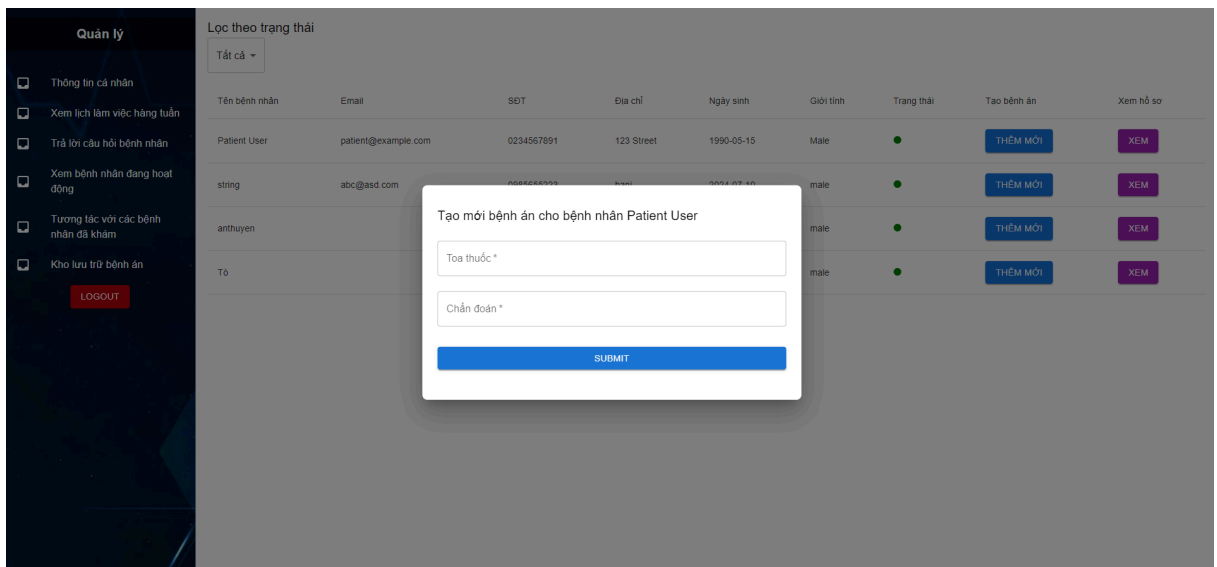


Figure 3.3.1.1 Create Medical Notebook

Function Details:

Normal cases

- Doctor logging into the system, then select “Thêm mới” to create new Medical Notebook, then fill all the fields and submit.

Abnormal cases

- If there is a field that is empty, the system will notify you.

3.3.2 Search Medical Notebook

Function Trigger:

- User input phone number of patient record into the search box and click the “Tìm kiếm” button.

Function Description:

- This function is used to search all of the Medical Notebooks.

Screen Layout:

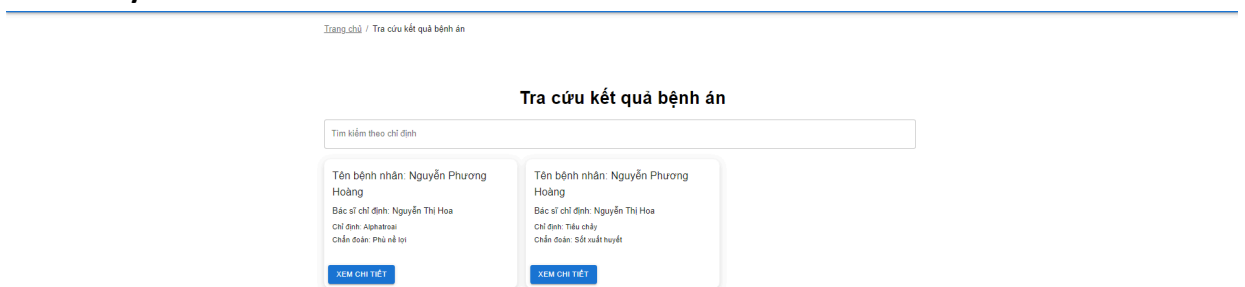


Figure 3.3.2.1 Search Medical Notebook

Function Details:

Normal cases

- User input phone number of patient record into the search box and click the “Tìm kiếm” button and view the all records.

Abnormal cases

- User input wrong phone number of patient but still click on “Tìm kiếm” button and the system will show the Medical Notebook by phone number user input.

3.3.3 View Medical Notebook

Function Trigger:

- The User clicks the button “View Medical Notebook” .

Function Description:

- This function Allows view the Medical Notebooks of patients.

Screen Layout:

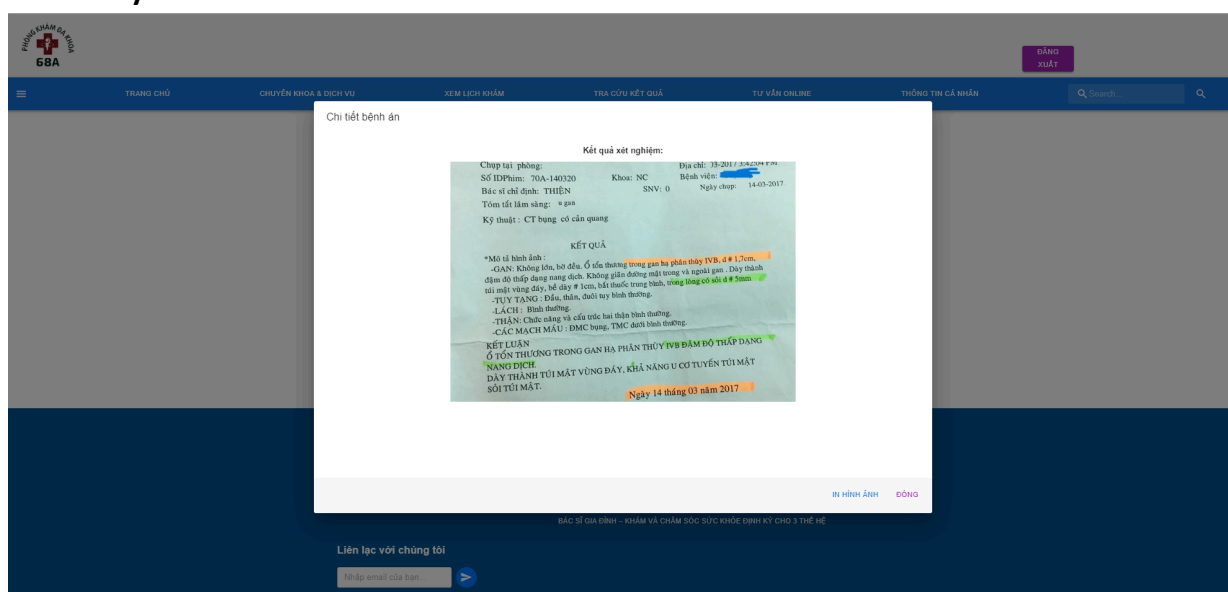


Figure 3.3.3.1 View Medical Notebook

Function Details:

Normal cases

- The system displays the Medical Notebook of the patient, showing patient information such as name, ID, medical condition...

Abnormal cases

- None.

3.3.4 Active patient status

Function Trigger:

- When the user clicks on click on “Tài khoản bệnh nhân ”

Function Description:

- User click on “Tài khoản bệnh nhân” to active the status of patient

Screen Layout:

ID	Số điện thoại	Email	Mật khẩu	Tên	Giới tính	Địa chỉ	Ngày sinh	Trạng thái hoạt động	Tạo lịch tái khám	Trạng thái
59	0862526888	Hoangnp@gmail.com	Hoangnp123	Nguyễn Phương Hoàng	Male	Km29 Làng Hòa Lạc	1992-12-12T00:00:00	Active	+ TÀI KHẨM	HOẠT ĐỘNG
60	0962504407	Dungth@gmail.com	Dungth123	Lê Trung Dũng	Male	PH Luxury	1985-12-12T00:00:00	Active	+ TÀI KHẨM	HOẠT ĐỘNG
61	0918686236	Huyntk@gmail.com	Huyntk123	Ngô Kiến Huy	Male	Điện Phương Hà Nội	1999-12-12T00:00:00	Active	+ TÀI KHẨM	HOẠT ĐỘNG
62	0981627016	linhd@gmail.com	linhd123	Chu Diệu Linh	Female	Sóc Sơn Hà Nội	2001-01-17T00:00:00	Active	+ TÀI KHẨM	KHÔNG HOẠT ĐỘNG
63	0877481211	Sangnv@gmail.com	Sangnv123	Nguyễn Văn Sang	Male	Hà Đức Hà Nội	1985-06-12T00:00:00	Active	+ TÀI KHẨM	KHÔNG HOẠT ĐỘNG

Figure 3.3.4.1 Active patient status

Function Details:

Normal cases

- User active the status for patient and the system change the status of patient's account

Abnormal cases

- none

3.3.5 Import result image

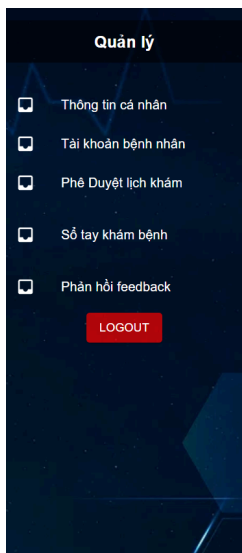
Function Trigger:

- When the receptionist clicks on the “Sổ tay khám bệnh” section and clicks on add icon to import image.

Function Description:

- Receptionists import images of medical notebooks for patients.

Screen Layout:



Kế đơn	Chẩn đoán	Tên bệnh nhân	Bác sĩ khám	Thêm ảnh	Lưu
Alphatroai	Phù nề lợi	Nguyễn Phương Hoàng	Nguyễn Thị Hoa	+	<button>SAVE</button>
Tiểu chảy	Sốt xuất huyết	Nguyễn Phương Hoàng	Nguyễn Thị Hoa	+	<button>SAVE</button>



c

Figure 3.3.6.1 Import result image

Function Details:

Normal cases

- Receptionist clicks on the add icon and chooses the image to import.

Abnormal cases

- none

3.3.6 View patient information

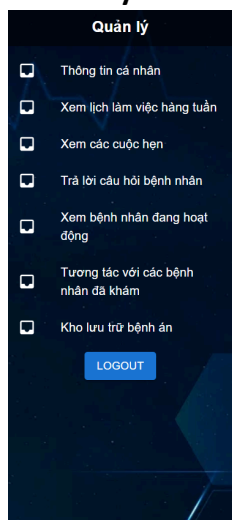
Function Trigger:

- When the doctor clicks on click on “xem bệnh nhân đang hoạt động”

Function Description:

- Doctor views patient information.

Screen Layout:



Lọc theo trạng thái									
Tất cả ▾									
Tên bệnh nhân	Email	SĐT	Địa chỉ	Ngày sinh	Giới tính	Trạng thái	Tạo bệnh án	Xem hồ sơ	
Nguyễn Phương Hoàng	Hoangnp@gmail.com	0862526888	Km29 Làng Hòa Lạc	1992-12-12	Male	●	<button>THÊM MỚI</button>	<button>XEM</button>	
Lê Trung Dũng	Dunght@gmail.com	0982504407	PH Luxury	1985-12-12	Male	●	<button>THÊM MỚI</button>	<button>XEM</button>	
Ngô Kiến Huy	Huynk@gmail.com	0918868236	Đan Phượng Hà Nội	1999-12-12	Male	●	<button>THÊM MỚI</button>	<button>XEM</button>	
Chu Diệu Linh	linhcd@gmail.com	0981627018	Sóc Sơn Hà Nội	2001-01-17	Female	●	<button>THÊM MỚI</button>	<button>XEM</button>	
Nguyễn Văn Sang	Sangnv@gmail.com	0877481211	Hòa Đức Hà Nội	1985-06-12	Male	●	<button>THÊM MỚI</button>	<button>XEM</button>	
Lưu Phương Chi	Chipl@gmail.com	0368631218	Sơn Tây Hà Nội	1986-12-01	Female	●	<button>THÊM MỚI</button>	<button>XEM</button>	

Figure 3.3.7.1 View patient information

Function Details:

Normal cases

- Doctor clicks on click on “xem bệnh nhân đang hoạt động” section to view patient information

Abnormal cases

- none

3.4 Marketing

3.4.1 View Customer feedback

Function Trigger:

- Customers can click on "Khách hàng nói gì về 68A" to see reviews from people who have used the service at the clinic.

Function Description:

- User can view reviews of previous customers through the system's reviews section

Screen Layout:



Figure 3.4.1.1 View Customer feedback

Function Details:

Normal cases

- Display the feedback of customers who have used the service at the clinic.

Abnormal cases

- Feedback will help users to briefly evaluate the quality of the clinic's service. From there, you can make a decision whether to use the clinic's services or not

3.4.2 Reply Customer feedback(receptionist)

Function Trigger:

- receptionist clicks on the text bar to write feedback of customer

Function Description:

- this function help the receptionist can reply the customer's feedback

Screen Layout:

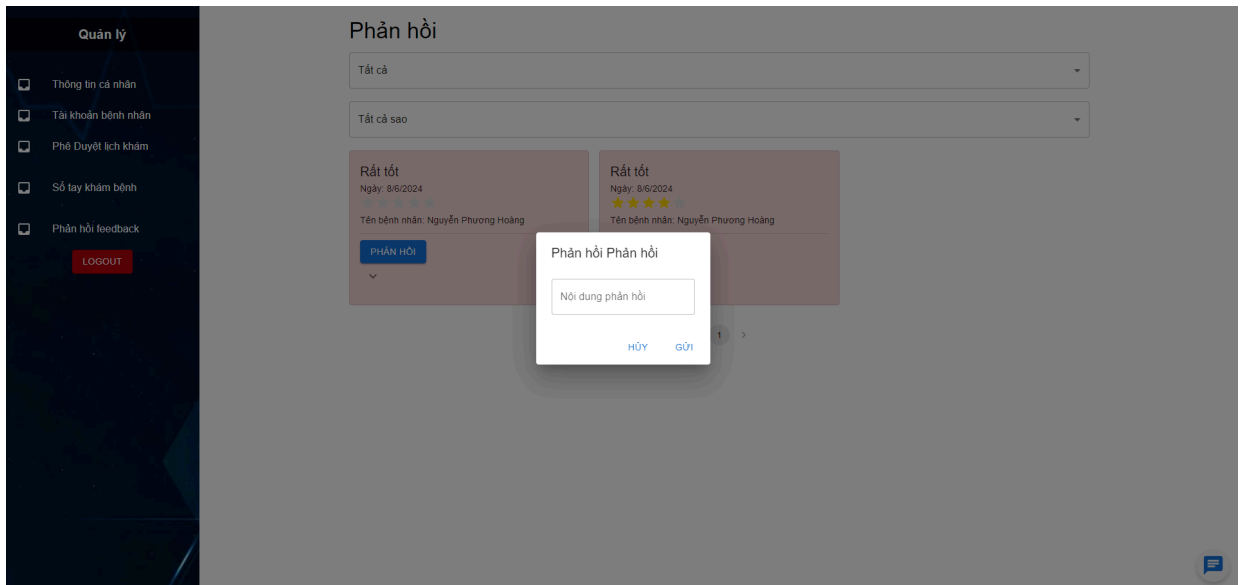


Figure 3.4.2.1 Reply Customer feedback(receptionist)

Function Details:

Normal cases

- Display the feedback of customers with text bar below for receptionist can reply the feedback

Abnormal cases

- the receptionist can reply without attach image

3.4.3 Create Customer feedback

Function Trigger:

- The system will send a feedback request when the patient completes the most recent examination schedule immediately after the patient logs in

Function Description:

- Users can leave feedback about the clinic after using the clinic service.

Screen Layout:

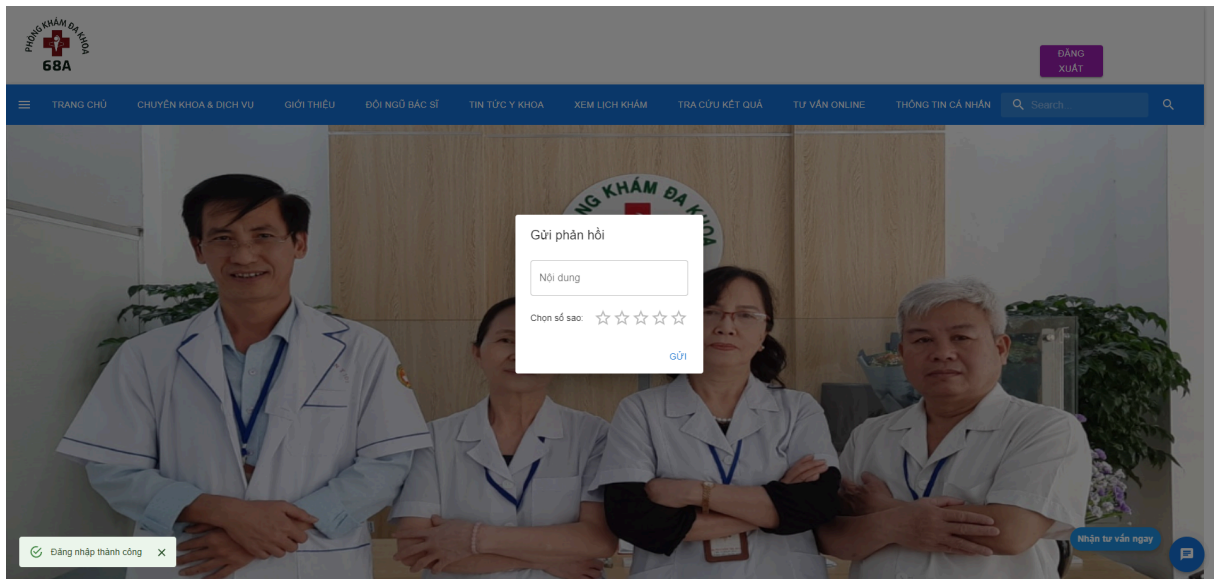


Figure 3.4.3.1 Create Customer feedback

Function Details:

Normal cases

- Users fill their feedback on the feedback box and attach images if necessary.

Abnormal cases

- none

3.4.4 View Medical information blog

Function Trigger:

- Users click on “tin tức y khoa ” button on the navbar to access the medical news section

Function Description:

- For users to access medical information about common diseases.

Screen Layout:

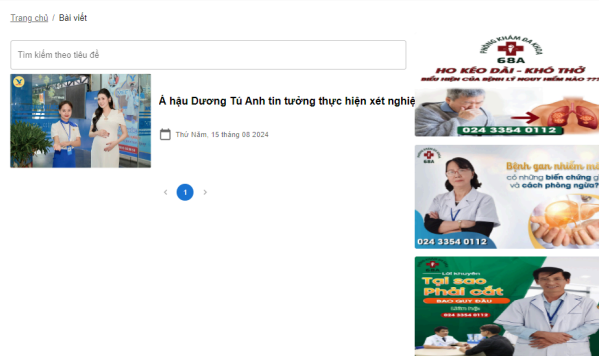


Figure 3.4.4.1 View Medical information blog

Function Details:

Normal cases

- Displays basic, verified medical information.

Abnormal cases

- Users will be provided with accurate medical information, how to prevent minor diseases and symptoms of the disease.

3.4.5 Create Medical information blog

Function Trigger:

- The Article manager clicks on the “Thêm mới ” button to create new the medical information blog.

Function Description:

- For the Article manager to create new medical information about common diseases.

Screen Layout:

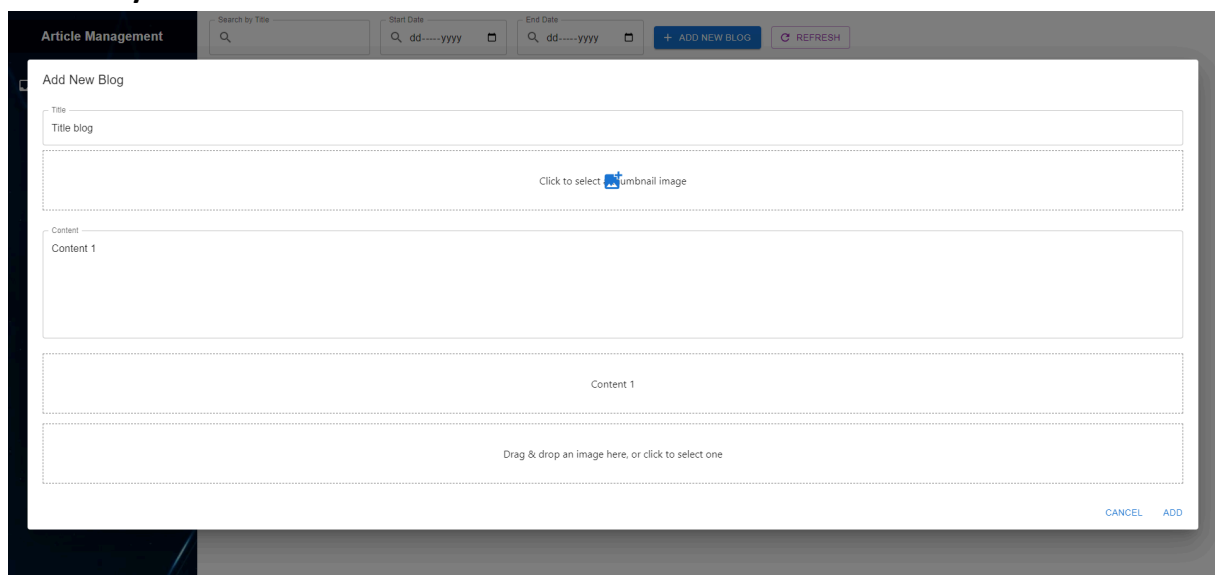
The screenshot shows a web application interface for 'Article Management'. At the top, there's a header with a search bar and filters for 'Start Date' and 'End Date', both set to 'dd----yyyy'. To the right of the filters are two buttons: '+ ADD NEW BLOG' and 'REFRESH'. Below the header is a modal window titled 'Add New Blog'. Inside the modal, there are four input fields: 'Title' (with placeholder 'Title blog'), 'Thumbnail image' (with placeholder 'Click to select thumbnail image'), 'Content' (with placeholder 'Content 1'), and another 'Content' field (also with placeholder 'Content 1'). At the bottom of the modal, there's a large area with the placeholder 'Drag & drop an image here, or click to select one'. In the bottom right corner of the modal, there are two buttons: 'CANCEL' and 'ADD'.

Figure 3.4.5.1 Create Medical information blog

Function Details:

Normal cases

- Display all fields initially as empty and require the content creator to fill all fields and select “Đăng bài” to create a medical information blog.

Abnormal cases

- The Article manager doesn't change any fields but selects the “Cập nhật” button.

3.4.6 Update Medical information blog

Function Trigger:

- The Article manager clicks on the “Cập nhật ” button to update the medical information blog.

Screen layout:

The screenshot shows a web application interface for 'Article Management'. At the top, there's a navigation bar with a search bar and date filters. Below this, a modal window titled 'Edit Blog' is open. Inside the modal, there are several input fields: a 'Title' field with the placeholder 'Blog Title', a large text area for 'Content' with the placeholder 'Content of the blog', and a thumbnail image selection area with the text 'Click to select thumbnail image'. Below the content area, there's a section for 'Blog Title' and 'Content of the blog' with a 'Drag & drop an image here, or click to select one' instruction. At the bottom right of the modal, there are 'CANCEL' and 'UPDATE' buttons.

Figure 3.4.6.1 Update Medical information blog

Function Description:

- For the Article manager to update medical information about common diseases.

Function Details:

Normal cases

- Display all old data information like title, paragraph and the Article manager selects the “Cập nhật” button to update the blog to the system.

Abnormal cases

- The Article manager doesn't change any fields but selects the “Cập nhật” button.

3.4.7 Delete Medical information blog

Function Trigger:

- The Article manager clicks on the “Xóa” button to remove the medical information blog.

Screen layout:

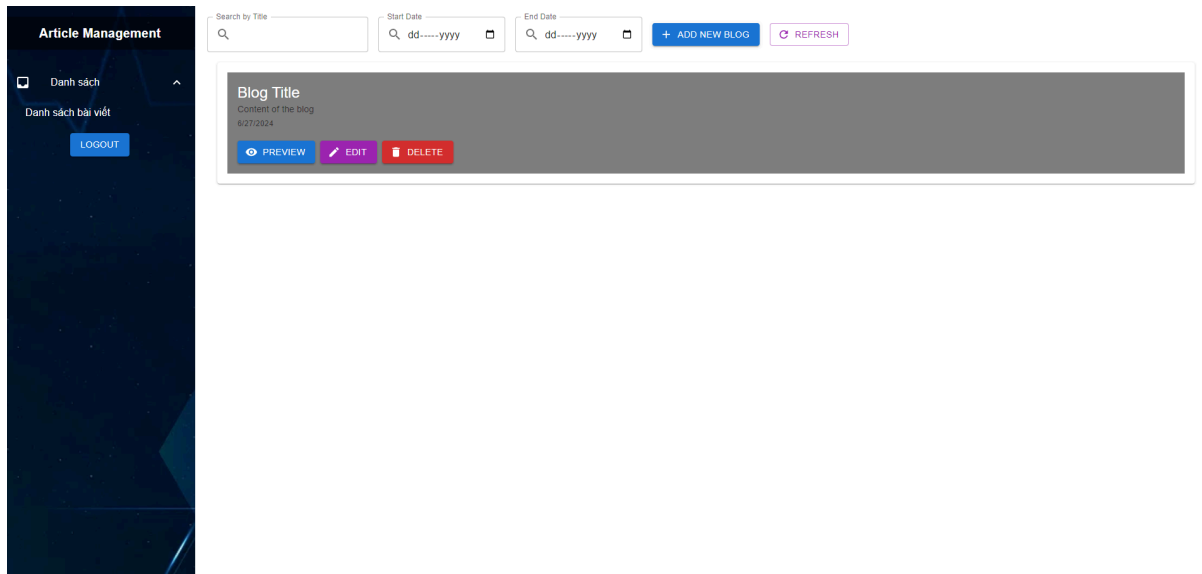


Figure 3.4.7.1 Delete Medical information blog

Function Description:

- For the Article manager to remove medical information about common diseases.

Function Details:

Normal cases

- Display all old data information like title, paragraph and the Article manager selects the “Xóa” button to confirm the removal of the blog to the system.

Abnormal cases

- The Article manager select “Hủy bỏ” button to cancel the removal of the blog.

3.4.8 Create Question for Doctor

Function Trigger:

- User clicks on “Đặt câu hỏi” button and fill the question form and clicks on “Gửi câu hỏi” button to submit a question for doctor, patient can also attach image if necessary.

Function Description:

- This function for patient ask doctor a question

Screen Layout:

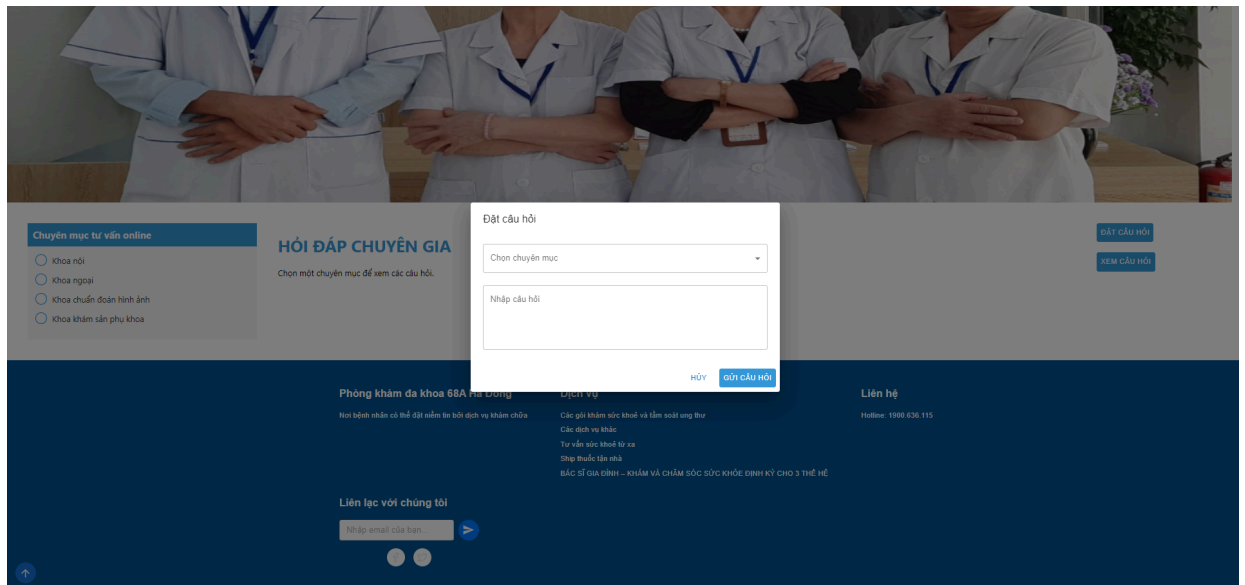


Figure 3.4.8.1 Create Question for Doctor

Function Details:

Normal cases

- User clicks on “Đặt câu hỏi” button and fill the question form, attach image and clicks on “Gửi câu hỏi” button to submit a question for doctor.

Abnormal cases

- User clicks on “Đặt câu hỏi” button and fill the question form without attach image and clicks on “Gửi câu hỏi” button and system still can submit a question for doctor.

3.4.9 Answer Question

Function Trigger:

- Doctor clicks on “Các câu hỏi” to view list of question of patient. Doctor clicks on the “Trả lời” button, fills the answer textbox or attaches an image and clicks “Lưu” to answer a question.

Function Description:

- This function is used to answer question of patient

Screen Layout:

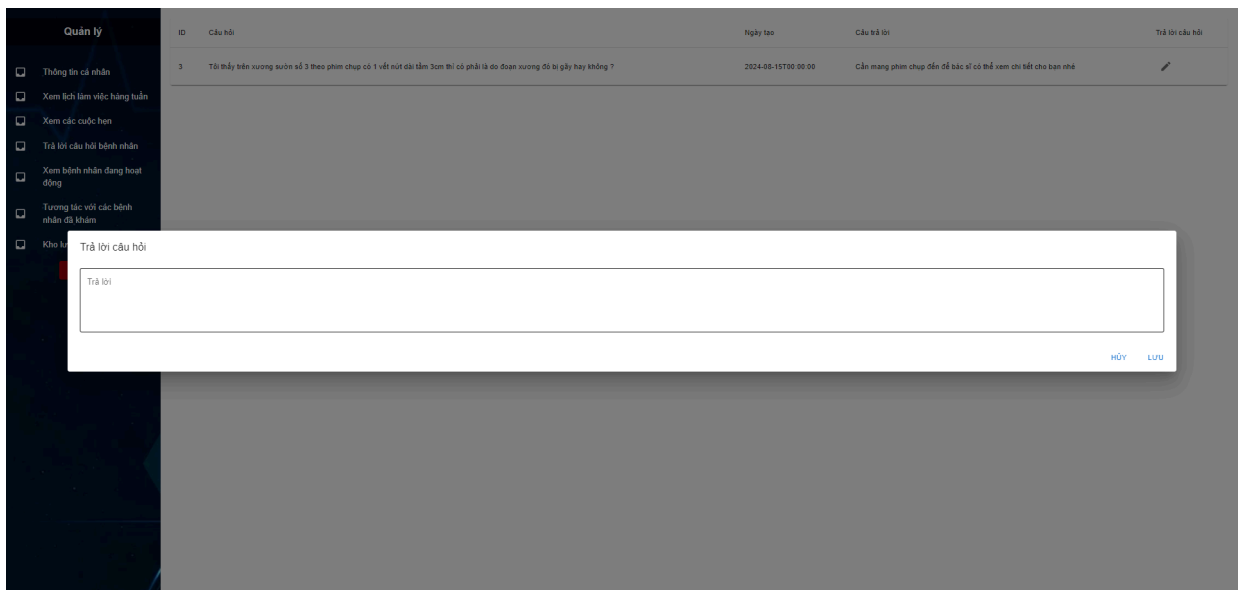


Figure 3.4.9.1 Answer Question

Function Details:

Normal cases

- User fill the answer and click “Gửi câu trả lời” to answer the question.

Abnormal cases

- Users who fill the textbox without an attached image still can answer the question.

3.4.10 View List of Blog Groups

Function Trigger:

- When the user clicks on “Tin tức y khoa” on the Navbar and choose option to view the blog groups

Function Description:

- The user enters the blog and chooses the option, the system retrieves and displays a list of available blog groups.

Screen Layout:

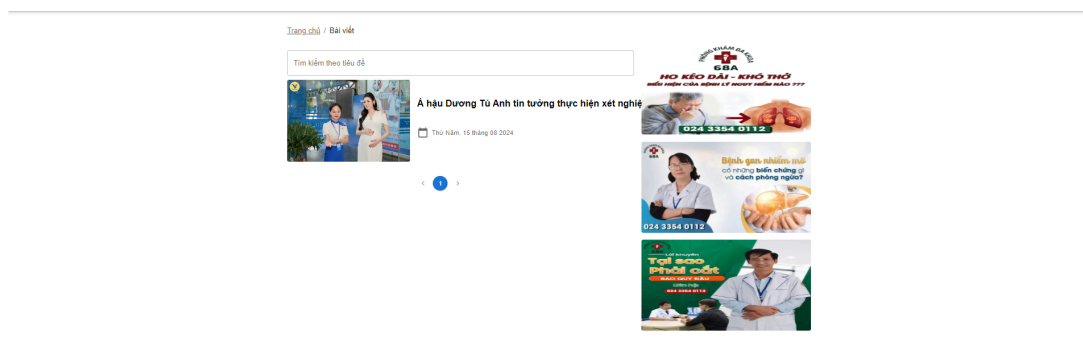


Figure 3.4.10.1 View List of Blog Groups

Function Details:

Normal cases

- system retrieves and displays a list of available blog groups.

Abnormal cases

- If the patient uses search or filter options to find specific blog groups, the system refines the list based on the criteria.

3.4.11 Search Blog

Function Trigger:

- When the user clicks on “Tin tức y khoa” on the Navbar and enter the keyword into the search bar to search the result they need

Function Description:

- The user enters the blog and enters the keyword or selects the filter into the search bar, the system retrieves and displays a list of available blog groups.

Screen Layout:

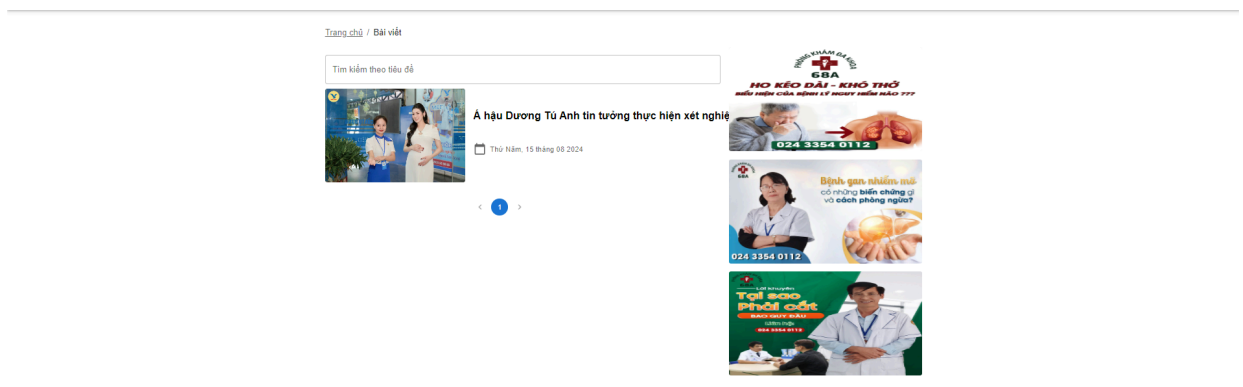


Figure 3.4.11.1 Search Blog

Function Details:

Normal cases

- The system processes the search query and retrieves blog posts that match the search criteria and displays a list of blog posts that match the search criteria.

Abnormal cases

- none

3.4.12 View service detail

Function Trigger:

- When the user clicks on a specific service on the home page to view details of service.

Function Description:

- The user enter the service detail and the system will show detail description of service the user want to view

Screen Layout:

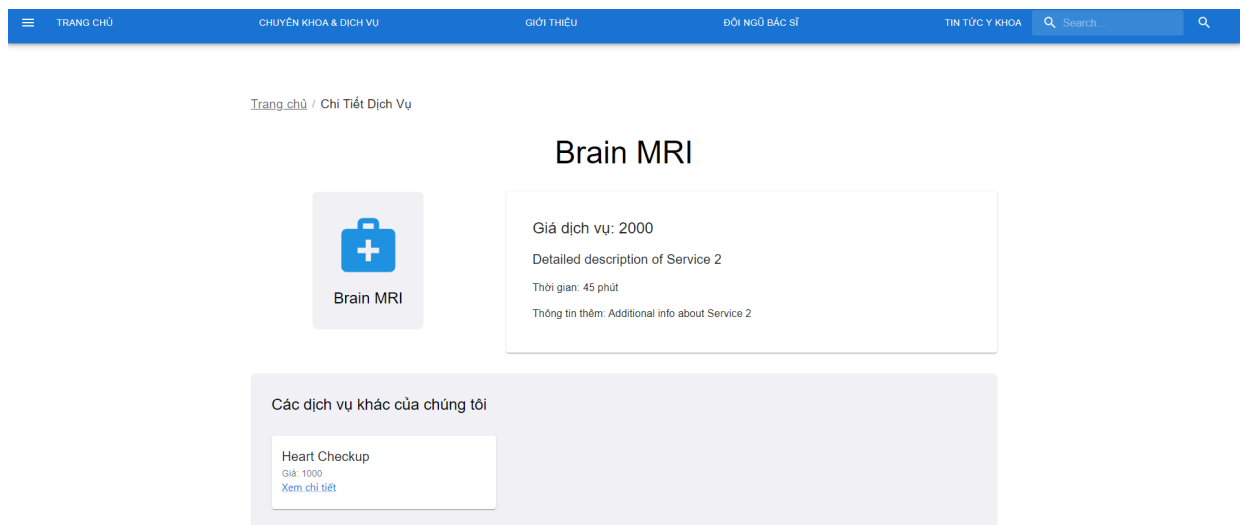


Figure 3.4.12.1 View service detail

Function Details:

Normal cases

- the user click on the service they want and the system show detail description of service the user want to view

Abnormal cases

- none

3.4.13 View List of Services

Function Trigger:

- When the user views the list of services on the “Các khoa y tế” section on the home page.

Function Description:

- The user views all the services on the polyclinic.

Screen Layout:

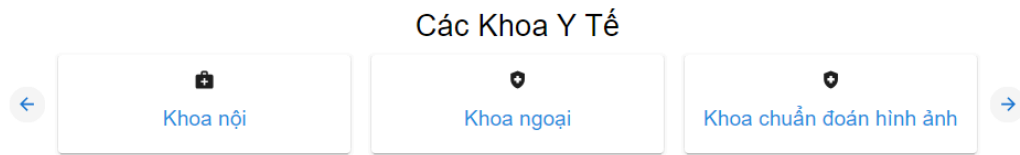


Figure 3.4.13.1 View List of Services

Function Details:

Normal cases

- The user click on the service and the system show list service of clinic

Abnormal cases

- None

3.4.14 View Doctor's Information

Function Trigger:

- When the user clicks on “ĐỘI NGŨ BÁC SĨ” on the Navbar or click on “Xem thêm” on ĐỘI NGŨ BÁC SĨ component.

Function Description:

- The user view list all of the doctors in the poly clinic and views the information of each doctor.

Screen Layout:

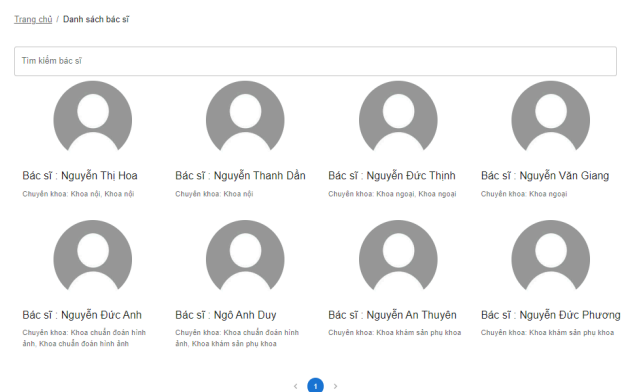


Figure 3.4.14.1 View Doctor's Information

Function Details:

Normal cases

- The user view list all of the doctors in the poly clinic and views the information of each doctor.

Abnormal cases

- *none*

3.4.15 View List of Doctors

Function Trigger:

- When the user clicks on “ĐỘI NGŨ BÁC SĨ” on the Navbar or click on “Xem thêm” on ĐỘI NGŨ BÁC SĨ component.

Function Description:

- The patient click on the doctor list on the Navbar and the system will show the list of doctor in clinic

Screen Layout:

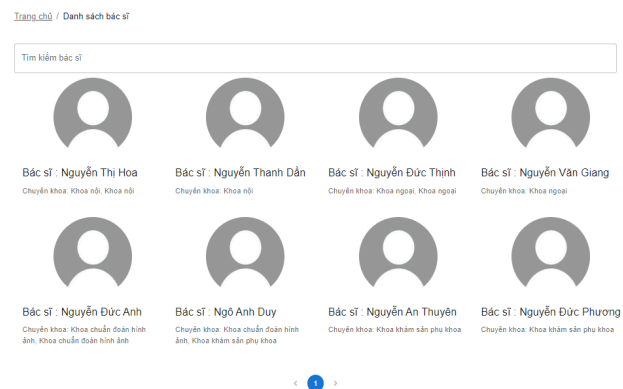


Figure 3.4.15.1 View List of Doctors

Function Details:

Normal cases

- The patient click on the doctor list on the Navbar and the system will show the list of doctor in clinic

Abnormal cases

- *none*

3.4.16 Update answer

Function Trigger:

- When the user clicks on “Trả lời câu hỏi bệnh nhân” on the sidebar to view list of question and click on edit icon to update their answer.

Function Description:

- The doctor click on the question and they can update their answer

Screen Layout:

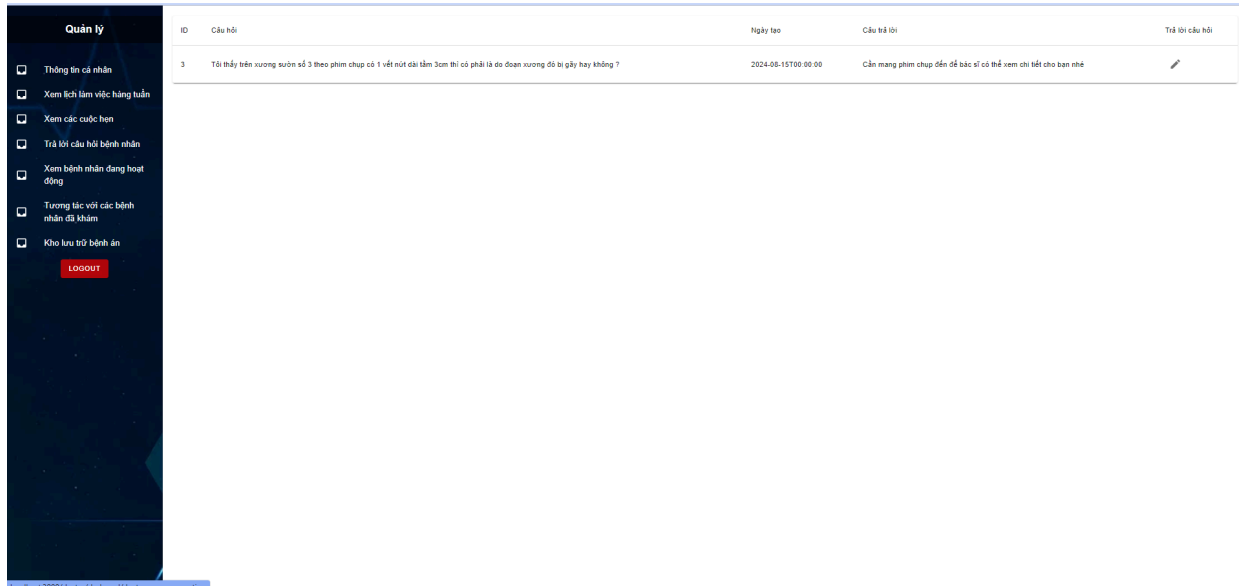


Figure 3.4.16.1 Update answer

Function Details:

Normal cases

- When the user clicks on “Trả lời câu hỏi bệnh nhân” on the sidebar to view list of question and click on edit icon to update their answer.

Abnormal cases

- none

3.4.17 Update question

Function Trigger:

- When the user clicks on “Hỏi Đáp chuyên gia ”section.
- User click on “Xem câu hỏi” button to view all the questions that they ask.

Function Description:

- The patient click on the question and they can update their question

Screen Layout:

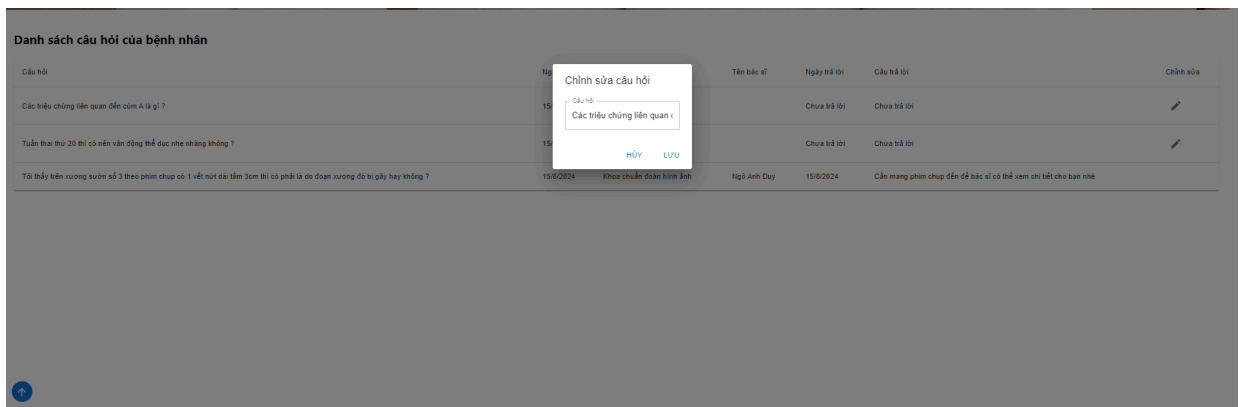


Figure 3.4.17.1 Update question

Function Details:

Normal cases

- When the user clicks on “Hỏi Đáp chuyên gia ” section.
- User click on “Xem câu hỏi” button to view all the questions that they ask.

Abnormal cases

- none

3.5. User and Access Management

3.5.1 Login

Function Trigger:

- When the user clicks on “Login” on the header.

Function Description:

- The user enters the account's email with password or password with password to log in to the system.

Screen Layout:

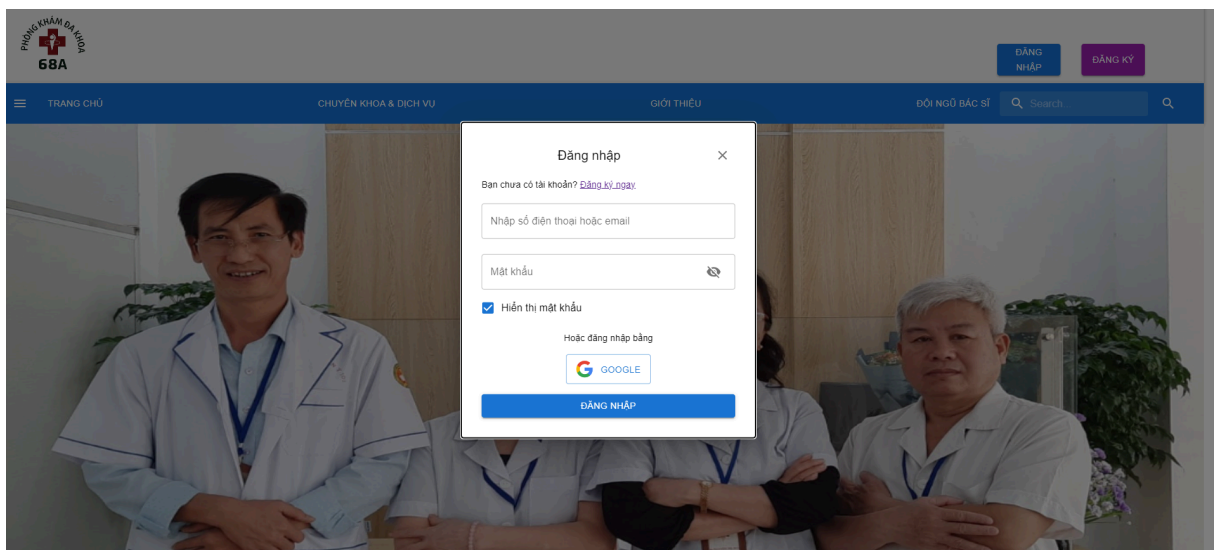


Figure 3.5.1.1 Login

Function Details:

Normal cases

- Systems display the login form for users to enter the email and password.
- The system will check the login information with the saved data.
- Valid credentials, the system will redirect to the homepage screen corresponding to that person's role

Abnormal cases

Invalid Credential:

- If the credentials are not correct, the System will provide instructions and allow the user to login again.

Invalid active:

- If the account doesn't have permission for user access, go to an unauthorised page.

3.5.2 Forgot password

Function Trigger:

- Director clicks on the “Quên mật khẩu” on login form and input phone number to verify OTP to get new password.

Function Description:

- The purpose of this use case is to allow the Director to set a new password for an account in case they lost their password.

Screen Layout:



Figure 3.5.2.1 Forgot password

Function Details:

1. **Normal cases**

- The system shows the reset password form for the Director including the new password and reenters the new password.
- User enters a new password and reenters the new password.
- System validates the new password and updates it for the user's account.
- System confirms the successful password reset to the Director.

2. **Abnormal cases**

- *Invalid password* (violates the BR01 of the system : Passwords must have at least 6 characters, including at least one number and one special character.): Systems notify: "Passwords must have at least 6 characters, including at least one number and one special character. please re-enter the password".
- *Users enter the re-enter password isn't the same as the password*: Systems notify: "The re-enter password is not the same as the password. please re-enter the password".

3.5.3 Logout

Function Trigger:

- If the user is already logged in, they can log out of their account.

Function Description:

- The process and associated interactions when a user initiates the logout action in an MedPal system.

Screen Layout:

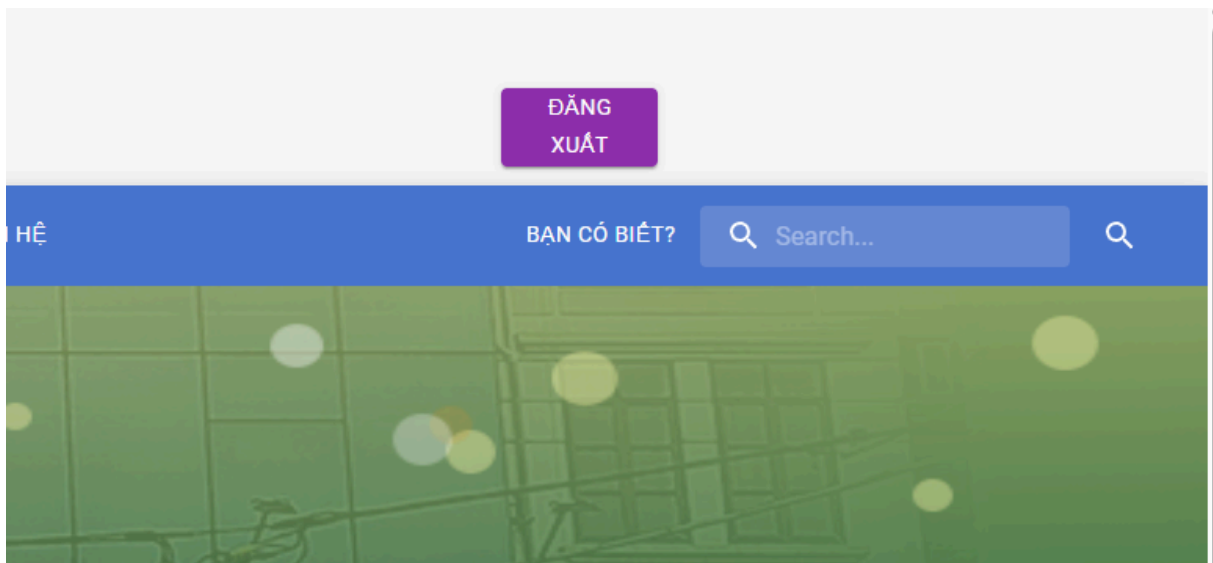


Figure 3.5.3.1 Logout

Function Details:

Normal cases

- When the user logged in and wants to log out, they can select “Đăng xuất” to log out, finishing the login session.

Abnormal cases

- None.

3.5.4 Register

Function Trigger:

- The user selects “Đăng ký” to register a new account.

Function Description:

- The patient decides to register for an account to access personalised Medpal services.

Screen Layout:

Xác Minh Thành Công ✕

Bạn đã xác minh OTP thành công. Vui lòng nhập thông tin cá nhân để hoàn tất quá trình.

Tên *

Số điện thoại: 0237264235

Email *

Ngày sinh *
mm/dd/yyyy

Giới tính *

Địa chỉ *

Mật khẩu *

Nhập lại mật khẩu *

HOÀN TẤT ĐĂNG KÝ

Figure 3.5.4.1 Register

Function Details:

Normal cases

- The user can access the Medpal registration page and fill in all necessary information, and select “Đăng ký” to confirm access to the system, if all fields are valid, a new account will be created.

Abnormal cases

- If there is a field that is not valid, a message will be delivered as a failure.

3.5.5 Update profile(all of actor)

Function Trigger:

- The user selects “Đăng ký” to register a new account.

Function Description:

- This use case allows users to update information about their accounts such as passwords, and personal information.

Screen Layout:

Figure 3.5.5.1 Update profile(all of actor)

Figure 3.5.5.2 Update profile(all of actor)

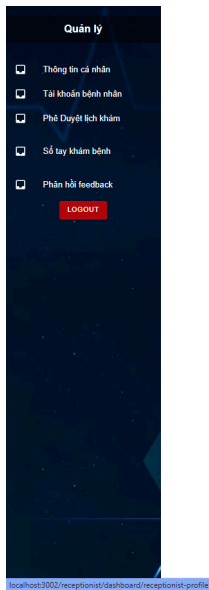


Figure 3.5.5.3 Update profile(all of actor)

Function Details:

Normal cases

- The user selects the profile icon and will be redirected to the profile, from here they can change some information fields such as passwords, etc.

Abnormal cases

- They can't update their profile if it is not valid (for example: If the phone number is invalid, a message will be delivered).

3.5.6 Create patient account(role receptionist)

Function Trigger:

- The receptionist chooses "Tài khoản bệnh nhân" on the sidebar.
- The receptionist clicks on the button "Thêm tài khoản" and inputs patient information to create a patient account.

Function Description:

- This use case allows the receptionist to provide an account to the patient.

Screen Layout:

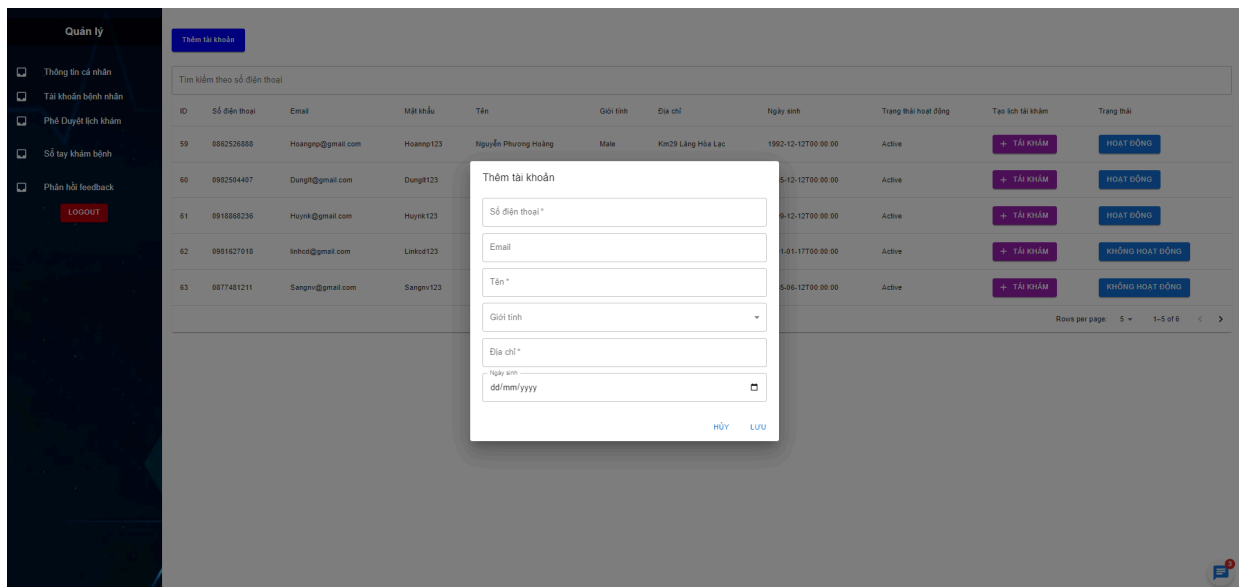


Figure 3.5.6.1 Create patient account(role receptionist)

Function Details:

Normal cases

- The receptionist clicks the “Tạo tài khoản bệnh nhân” button then the system will redirect the user to create page to enter a new patient .Then The system validates the patient input information.If the information is valid,a new account will be created for the patients.

Abnormal cases

- If the user does not enter required information fields such as name, code... The system will notify you of missing information and ask the user to enter all required information to create a new request for examination.

3.5.7 Create employee account

Function Trigger:

- User clicks on “Thêm tài khoản” and fills all the fields with employee information then clicks on the “Tạo” button to create a new employee.

Function Description:

- The purpose of this function is to create a new profile for employees.

Screen Layout:

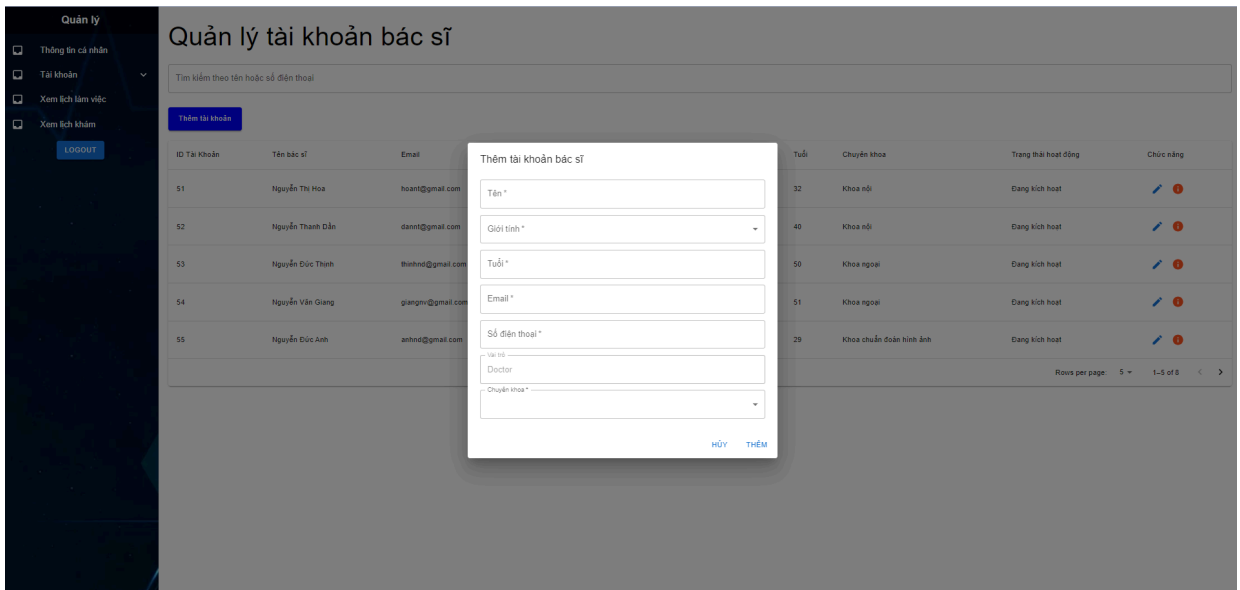


Figure 3.5.7.1 Create employee account

Function Details:

Normal cases

- The administrator clicks on the "Add account" button and a dialog appears to enter the user's data. If the data is in the correct format, a new employee account will be added

Abnormal cases

- If the user does not enter required information fields such as name, code... The system will notify you of missing information and ask the user to enter all required information to create a new request for examination.

3.5.8 Change status of employee account(actor admin)

Function Trigger:

- User click on the edit icon to change the status of the employee they quit.

Function Description:

- For user to change employee status.

Screen Layout:

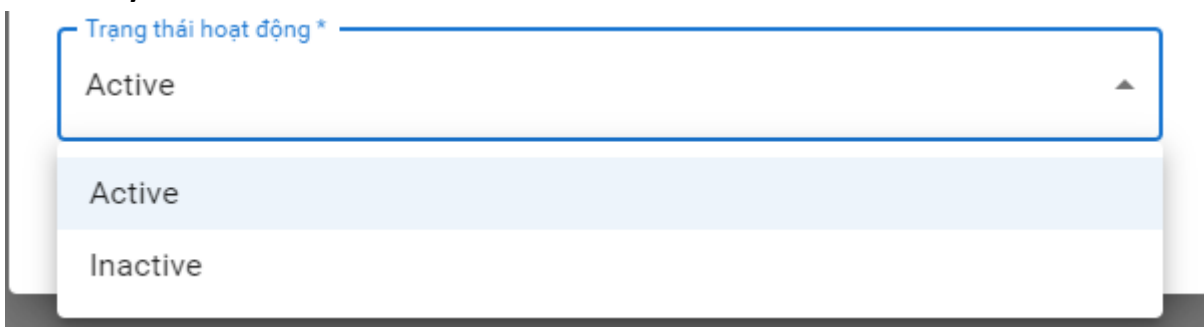


Figure 3.5.8.1 Change status of employee account(actor admin)

Function Details:

Normal cases

- User clicks on the edit icon and changes the status of the employee.

Abnormal cases

- If user clicks on “Hủy”, the profile will not remove.

3.5.9 View list employee profile

Function Trigger:

- User clicks on the section “ Tài khoản” on the sidebar to view the list of employee profiles.

Function Description:

- To view the employee profile.

Screen Layout:

ID Tài Khoản	Tên bác sĩ	Email	Số điện thoại	Mật khẩu	Giới tính	Tuổi	Chuyên khoa	Trạng thái hoạt động	Chức năng
51	Nguyễn Thị Hoa	hoant@gmail.com	0328815402	Hoant123	Female	32	Khoa nội	Đang kích hoạt	 
52	Nguyễn Thanh Dân	dannt@gmail.com	0965259433	Dannt123	Male	40	Khoa nội	Đang kích hoạt	 
53	Nguyễn Đức Thịnh	thinhnd@gmail.com	0913552099	Thinhnd123	Male	50	Khoa ngoại	Đang kích hoạt	 
54	Nguyễn Văn Giang	giangnv@gmail.com	0986526526	Giangnv123	Male	51	Khoa ngoại	Đang kích hoạt	 
55	Nguyễn Đức Anh	anhnd@gmail.com	0985655223	anhnd123	Male	29	Khoa chuẩn đoán hình ảnh	Đang kích hoạt	 

Figure 3.5.9.1 View list employee profile

Function Details:

Normal cases

- User view profile of employee.

Abnormal cases

- The profile doesn't exist in the database and will not show.

4. Non-Functional Requirements

4.1 External Interfaces

The system should have interfaces to communicate with users through various channels such as web interfaces, mobile applications, email, and SMS.

4.2 Quality Attributes

4.2.1 Usability

- The system should have an intuitive user interface that requires minimal training for both normal users and power users.

- The average time for a user to book an appointment should not exceed X minutes.

4.2.2 Reliability

- The system should have an availability of at least 99.9% during business hours.
- The system's output should have an accuracy level of at least 99%.

4.2.3 Performance

- The average response time for booking an appointment should not exceed 3 seconds.
- The system should be capable of handling a peak load of at least 120 appointments per minute.
- The system should have a capacity to accommodate at least 500 concurrent users.
- Resource utilization should be monitored and optimized to ensure efficient performance.

4.2.4 Availability

- The system should be available 24/7, with scheduled maintenance windows communicated to users in advance.
- In case of scheduled maintenance or downtime, users should be notified in advance and provided with alternative means to book appointments.

4.2.5 Security

- Authentication and authorization: The software should provide secure user authentication and role-based access control.
- Data protection: The system should ensure the confidentiality, integrity, and availability of sensitive data.

5. Requirement Appendix

5.1 Business Rules

Table 5.1.1 Business Rules

ID	Rule Definition
BR-01	Delivery time windows are 15 minutes, beginning on each quarter hour.
BR-02	Deliveries must be completed between 10:00 A.M. and 2:00 P.M. local time, inclusive.
BR-03	All meals in a single order must be delivered to the same location.
BR-04	All meals in a single order must be paid for by using the same payment method.
BR-11	If an order is to be delivered, the patron must pay by payroll deduction.
BR-12	Order price is calculated as the sum of each food item price times the quantity of that food item ordered, plus applicable sales tax, plus a delivery charge if a meal is delivered outside the free delivery zone.
BR-24	Only cafeteria employees who are designated as Menu Managers by the Cafeteria Manager can create, modify, or delete cafeteria menus.
BR-33	Network transmissions that involve financial information or personally identifiable information require 256-bit encryption.

BR-86	Only regular employees can register for payroll deduction for any company purchase.
BR-88	An employee can register for payroll deduction payment of cafeteria meals if no more than 40 percent of his gross pay is currently being deducted for other reasons.

5.2 System Messages

Table 5.2.1 System Messages

#	Message code	Message Type	Context	Content
1	MSG01	In line	There is not any search result	<i>No search results.</i>
2	MSG02	In red, under the text box	Input-required fields are empty	<i>The * field is required.</i>
3	MSG03	Toast message	Updating asset(s) information successfully	<i>Update asset(s) successfully.</i>
4	MSG04	Toast message	Adding new asset successfully	<i>Add assets successfully.</i>
5	MSG05	Toast message	Confirming email of asset hand-over is sent successfully	<i>A confirmation email has been sent to {email_address}.</i>
6	MSG06	Toast message	Resetting asset information successfully	<i>Return asset(s) successfully.</i>
7	MSG07	Toast message	Deleting asset information successfully	<i>Delete asset(s) successfully.</i>
8	MSG08	In red, under the text box	Input value length > max length	<i>Exceed max length of {max_length}.</i>
9	MSG09	In line	Username or password is not correct when clicking sign-in	<i>Incorrect username or password. Please check again.</i>